Path Integrals

Part III: Quantum Field Theory

The field path integral, Feynman diagrams and renormalization

Typeset from handwritten notes (pp. 1–25), corrected, reorganized and expanded

Completes the series begun in Part I (non-relativistic quantum mechanics) and Part II (relativistic quantum mechanics)

October 2026

Preface

These notes are a typeset version of handwritten lecture notes on the path integral for quantum field theory (handwritten pp. 1–25). They are Part III, the last part, of a series. Part I treats the path integral of non-relativistic quantum mechanics. Part II treats the relativistic particle, its worldline path integral and the free Klein–Gordon field. Chapter, equation and figure numbers continue those of Parts I and II: Chapters 1–9 are Part I, Chapters 10–18 are Part II, and this part begins at Chapter 19. A reference such as eq. (6.7) or §15.3 therefore points into an earlier part, and the appendices here continue from Appendix E.

The handwritten notes set two goals [hw. p. 1]. Goal I is an algorithm that produces the N-point functions $G^{(N)}$ of an interacting scalar field systematically from the field path integral. Goal II is to make sense of these functions quantitatively, which requires renormalization. The handwritten pages carry Goal I as far as the exponentiation of connected diagrams. This version completes Goal I and adds a self-contained introduction to Goal II, so the series ends with a working theory: Feynman rules derived, loops computed, infinities removed and the results connected to experiment.

Three kinds of changes were made, as in Parts I and II.

1. **Corrections.** The sign of the interaction terms in the Lagrangian, two stray factors of $\frac{1}{2}$, the argument that the vacuum is constant, the role of the vacuum energy in the projection onto the vacuum, and the claim that the vacuum sits at $\phi = 0$ have been fixed. The complete log, page by page, is in Appendix F.

2. **Reordering.** Two exactly solvable warm-up chapters now come before the general formalism, so that every combinatorial statement can be checked first on an ordinary integral and on a driven oscillator. The exponentiation of connected diagrams now comes before the cancellation of vacuum bubbles, which depends on it.

3. **Additions**, marked ★ in headings: field theory in zero dimensions; the driven oscillator; the $i\eta$ prescription as a rotation of time and the Euclidean theory; the Yukawa potential; the zero-point energy and the Casimir effect; the free field as a gas of worldlines; the interacting Feynman rules with worked examples; connected functions as cumulants and the vacuum energy; the Schwinger–Dyson equations; the effective action and the effective potential; vacuum structure beyond perturbation theory; power counting, one-loop renormalization and the running coupling; the spectral representation and the LSZ formula; and solutions to all exercises.

Tags such as [hw. p. 12] point to the handwritten QFT notes.

Conventions used throughout Part III (those of Part II, plus sources)

- Mostly-plus signature $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, \ldots, +1)$; D spacetime and $N = D - 1$ space dimensions; $c = 1$, $\hbar$ explicit. The field's mass parameter is the inverse Compton wavelength $\mu = m/\hbar$, and $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + \mu^2}$.
- Plane waves $e^{ik\cdot x}$ with $k \cdot x = -k^0 t + \mathbf{k} \cdot \mathbf{x}$. In spacetime $\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^{D/2}} e^{ik\cdot x} \tilde{\phi}(k)$ [hw. p. 15]; in space, as in Part II, $\phi(t, \mathbf{x}) = \int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{\phi}(t, \mathbf{k})$.
- The Lagrangian is $\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - U(\phi)$ with $U = \frac{1}{2}\mu^2\phi^2 + V_{\mathrm{int}}$ and $V_{\mathrm{int}} = \frac{g_3}{3!}\phi^3 + \frac{g_4}{4!}\phi^4$.
- $G_{\mathrm{F}}(x - y) = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ and $\Delta \equiv iG_{\mathrm{F}}$ (the handwritten notation); the regulator is $i\eta$; Euclidean time is $x^0 = -i\bar{x}^0$, and $G_{\mathrm{E}} = \hbar(-\bar{\partial}^2 + \mu^2)^{-1}$.
- Sources enter the action as $\hbar J\phi$, so the exponent is $\frac{i}{\hbar}S[\phi] + i(J, \phi)$ with $(J, \phi) \equiv \int \mathrm{d}^D x\, J(x)\phi(x)$.
- $\mathcal{Z}[J]$ is the unnormalized generating functional, $Z[J] = \mathcal{Z}[J]/\mathcal{Z}[0]$ generates all $G^{(N)}$, $W[J] = \ln Z[J]$ generates the connected ones, and $\Gamma[\Phi]$ is the effective action, $\frac{i}{\hbar}\Gamma = W - i(J, \Phi)$. (Srednicki's W is $-iW$ in this normalization.)

quantity	handwritten	here
generating functional of all $G^{(N)}$	$W[J]$	$Z[J]$
the same, normalized by the free vacuum	$\tilde{W}[J]$, $W_0[J]$	$\tilde{Z}[J]$, $Z_0[J]$
generating functional of connected diagrams	$Z[J]$, $\tilde{Z}[J]$	$W[J]$, $\tilde{W}[J]$
coupling constants	λ_3, λ_4	g_3, g_4
mass parameter on the field side	m	$\mu = m/\hbar$
wave vector	p	k
interaction potential (p. 20)	$V(\phi)$	$V_{\mathrm{int}}(\phi)$
numbers of vertices and propagators	V, P	v, p
spacetime integral	$\langle f\rangle$, $\langle J\phi\rangle$	$\int \mathrm{d}^D x\, f$, (J, ϕ)
classical solution (p. 4)	Φ	ϕ_{c}
regulator	$i\epsilon$	$i\eta$

The names Z and W are swapped relative to the handwritten notes so that they agree with Parts I and II, where Z is a partition function and $\ln Z$ its connected part (the vacuum loop of eq. (17.10)), and with the standard texts. The couplings are g_n because λ is the einbein of Part II.

What carries over from Parts I and II

- Projection onto the ground state by a tilted time, eq. (8.7) (§8.4): the vacuum Green's functions of Chapter 23.
- The Schwinger–Dyson equation, eq. (7.16), and contact terms (§7.7): Chapter 28.
- Imaginary time as statistical mechanics and the partition function (§8.7, §8.11): Euclidean field theory (§23.5) and the vacuum energy (§27.4).
- Stationary phase (§2.5, §4.6): the loop expansion (§20.5) and the effective action (Chapter 29).
- The harmonic oscillator (§9): the driven oscillator (Chapter 22) and the field as one oscillator per mode (§22.5).
- The free Klein–Gordon field and its Green's functions (§15): the free generating functional (Chapter 24).
- The equivalence of the worldline and the field, eq. (16.8), and the closed worldline, eq. (17.10): the gas of worldlines (Chapter 25) and the effective potential (§29.3).
- The diagrams and $\hbar$-counting of §17.8, which Part II stated without proof: derived in Chapter 26.
- The field path integral, eq. (18.9), and the vacuum wave functional, eq. (18.8): Chapters 19 and 23.

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19 The field path integral

19.1 The problem and the two goals [hw. p. 1]

Part II ended with a self-interacting scalar particle of mass m and couplings $g_3, g_4, \ldots$ (§17.9). Such a theory is specified by its N-point functions $G^{(N)}(x_1, \ldots, x_N)$, $N = 0, 1, 2, \ldots$, and these were computed by Feynman rules: draw the topologically distinct diagrams, let each line be a worldline path integral, and give each vertex a factor. The rules for the lines were derived there. The rules for the vertices, and the symmetry factors, were stated without proof [hw. p. 1].

The claim of the handwritten notes is that the same $G^{(N)}$ follow systematically from the path integral of the spacetime field theory. That sets two goals [hw. p. 1].

I. **An algorithm:** derive the Feynman rules for $G^{(N)}$ from the field path integral, with every factor fixed. Chapters 19–28 do this.

II. **Making sense of $G^{(N)}$ quantitatively.** Loop diagrams contain integrals that diverge at short distances; renormalization removes the divergences in favour of measured parameters. Chapters 29–31 introduce this and connect $G^{(N)}$ to experiment.

The starting point is Part I's exact statement [hw. pp. 1–2]. For $H = \frac{p^2}{2m} + U(x)$, with Lagrangian $L = \frac{1}{2}m\dot{x}^2 - U(x)$ and $S = \int \mathrm{d}t\, L$,

$$\langle x_f, t_f | \mathcal{T}[\cdots] | x_i, t_i \rangle = \mathcal{N} \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x\; e^{iS[x]/\hbar}\, [\cdots], \tag{19.1}$$

where $[\cdots]$ stands for insertions of the path at intermediate times (eq. (7.5)). The Hamiltonian defines the Schrödinger calculation on the left; the action defines the path integral on the right. They agree because H is quadratic in p (Case 1, §5.3).

19.2 From one coordinate to a field [hw. pp. 2–3]

The analogy of Part II's Additional Note 03 (§18.1) replaces the single coordinate x by one coordinate $\tilde{\phi}(\mathbf{k})$ for every wave vector $\mathbf{k}$, an uncountable set of degrees of freedom:

$$x \;\longleftrightarrow\; \tilde{\phi}(\mathbf{k}) \;\; \forall \mathbf{k}, \qquad |t, x\rangle \;\longleftrightarrow\; |t, [\tilde{\phi}]\rangle, \qquad \hat{\tilde{\phi}}(t, \mathbf{k})\, |t, [\tilde{\phi}]\rangle = \tilde{\phi}(\mathbf{k})\, |t, [\tilde{\phi}]\rangle. \tag{19.2}$$

Here $\hat{\tilde{\phi}}(t, \mathbf{k})$ is the Heisenberg operator of canonical quantization and $|t, [\tilde{\phi}]\rangle = e^{i\hat{H}t/\hbar} |[\tilde{\phi}]\rangle$ is its eigenstate, with Part I's convention for Heisenberg states (eq. (18.1)). The eigenvalue is a c-number function $\tilde{\phi}(\mathbf{k})$, and it is this function that will be integrated over. Part II (§18.2) noted that $\hat{\tilde{\phi}}(\mathbf{k})$ is not Hermitian, so its eigenvalues obey

$\tilde\phi(-\mathbf{k}) = \tilde\phi(\mathbf{k})^*$: they are the Fourier components of a real field. Transforming back,

$$\phi(t,\mathbf{x}) = \int \frac{\mathrm{d}^N k}{(2\pi)^{N/2}}\, \tilde\phi(t,\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \hat\phi(t,\mathbf{x})\,|t,[\phi]\rangle = \phi(\mathbf{x})\,|t,[\phi]\rangle \tag{19.3}$$

[hw. p. 2; the handwritten $\mathrm{d}p$ is $\mathrm{d}^N k$]. The labels $[\tilde\phi]$ and $[\phi]$ carry the same information [hw. p. 3]: the Fourier transform is unitary, so the states are the same and the two integration measures differ by a constant Jacobian.

19.3 The path-integral statement [hw. p. 3]

The derivation of Part I, Chapter 5, goes through for fields (§18.5). With insertions it gives the generalization of (19.1):

The field path integral

$$\langle t_f,[\phi_f]|\mathcal{T}[\cdots]|t_i,[\phi_i]\rangle = \mathcal{N}\int_{\phi(t_i)=\phi_i}^{\phi(t_f)=\phi_f} \mathcal{D}\phi\; e^{iS[\phi]/\hbar}\,[\cdots], \tag{19.4}$$

$$S[\phi] = \int \mathrm{d}^D x\,\mathcal{L}, \qquad \mathcal{L} = -\frac{1}{2}\,\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - U(\phi). \tag{19.5}$$

The boundary conditions hold at every point of space, $\phi(t_{i,f},\mathbf{x}) = \phi_{i,f}(\mathbf{x})$, and $[\cdots]$ is a product of fields at intermediate times on the right and the corresponding time-ordered operators on the left.

Why this form of $\mathcal{L}$. The handwritten notes stress that the statement holds for a Lagrangian of exactly this form [hw. pp. 3, 15]. The reason is Case 1 of Part I. The Hamiltonian density

$$\mathcal{H} = \frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\phi)^2 + U(\phi) \tag{19.6}$$

is quadratic in the momentum Π with a field-independent coefficient, so the momentum integrals of the phase-space path integral are Gaussian and produce exactly $e^{iS/\hbar}$ with the Lagrangian measure. With derivative interactions, such as $\phi^2\,\partial_\mu\phi\,\partial^\mu\phi$, the coefficient of Π^2 depends on ϕ; integrating out Π then leaves extra measure factors, and insertions of $\partial_0\phi$ produce contact terms (§7.7). Neither complication arises here.

19.4 Making it finite: the lattice [hw. p. 3]

The handwritten note on p. 3 says how to give (19.4) a meaning: put space on a lattice, so that the degrees of freedom become countable. Take a cubic lattice of spacing a in a box of side L. Then there are finitely many sites $\mathbf{x}_n$, and finitely many wave vectors, with components in $(2\pi/L)\mathbb{Z}$ and $|k_j| \le \pi/a$. Each site carries one coordinate $\phi_n(t)$,

which is a path in time; equivalently each discrete wave vector carries one [hw. p. 3]. The field path integral is then a multiple path integral of Part I, one for each site, and with time sliced as well it is an ordinary integral over finitely many variables.

For the free field the lattice Hamiltonian is a set of coupled oscillators, the mattress of Fig. 36:

$$H = \sum_n a^N \Big[\frac{1}{2}\Pi_n^2 + \sum_{j=1}^{N} \frac{(\phi_{n+e_j} - \phi_n)^2}{2a^2} + \frac{1}{2}\mu^2\phi_n^2 \Big], \tag{19.7}$$

where e_j steps to the neighbouring site in direction j. Its normal modes are plane waves with

$$\omega_{\mathbf{k}}^2 = \mu^2 + \sum_{j=1}^{N} \frac{4}{a^2} \sin^2 \frac{k_j a}{2} \quad \xrightarrow{ka \ll 1} \quad \mu^2 + \mathbf{k}^2. \tag{19.8}$$

Long wavelengths see the continuum dispersion relation; short ones see the lattice, and the frequencies stop growing at $\omega \sim 2\sqrt{N}/a$.

Two limits remain. $L \to \infty$ makes the wave vectors continuous; it is an infrared limit and causes no difficulty for a massive field. $a \to 0$ restores the continuum; it is an ultraviolet limit. In Part I the analogous limit, infinitely fine time slicing, was harmless. In field theory the number of degrees of freedom per unit volume grows like a^{-N}, and every one of them fluctuates. Zero-point energies (§24.4) and loop integrals (§26.4) depend on a. The lattice spacing is the first example of an ultraviolet cutoff, $\Lambda \sim \pi/a$, and controlling the dependence on it is Goal II (Chapter 30).

20 The vacuum and the shape of perturbation theory

20.1 Real time or Euclidean time [hw. p. 4]

The weight $e^{iS/\hbar}$ has modulus one: the integrand oscillates without decaying, exactly the problem of Part I, §8. There are two cures [hw. p. 4, comment 1]: rotate to Euclidean time, where the weight becomes $e^{-S_E/\hbar}$ (§8.8), or stay in real time with the $i\eta$ prescription (§8.9). For fields both are developed in §23.4 and §23.5, where they turn out to be the same thing, as they were in Part I.

20.2 The classical vacuum [hw. p. 4]

Matrix elements can be computed by expanding the action about a classical solution [hw. p. 4, comment 2]: this is stationary phase, the classical limit of §4.6. For the vacuum the relevant classical configuration is the one of lowest energy. By (19.6) the energy of a configuration is

$$E[\phi] = \int \mathrm{d}^N x \left[\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + U(\phi)\right]. \tag{20.1}$$

The first two terms are never negative and vanish only for a constant field. The energy is therefore lowest for a constant $\phi = \phi_0$ at the global minimum of U:

$$U'(\phi_0) = 0, \qquad U''(\phi_0) > 0. \tag{20.2}$$

This is the equilibrium condition of [hw. p. 4] (the handwritten ∂U means U').

> **Caution: why the vacuum is constant**
>
> The handwritten argument [hw. p. 4, comment 3] is that the vacuum must be Lorentz invariant, and that a Lorentz-invariant configuration $\phi_c(t, \mathbf{x})$ must be constant. The second step is false: any function of the invariant interval about a chosen origin, $\phi_c(x) = f(\eta_{\mu\nu}x^\mu x^\nu)$, is Lorentz invariant. What forces a constant is *translation* invariance, the requirement that the vacuum looks the same everywhere; or, as above, minimal energy. A constant scalar is then automatically Lorentz invariant. The same argument shows why only scalar fields can have a vacuum value: a constant vector A^μ would single out a direction in spacetime.

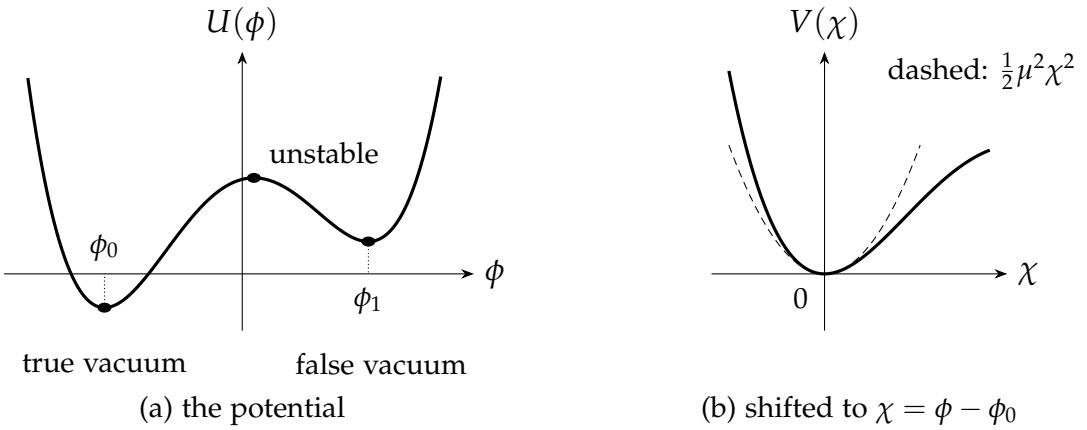
Stability. Expand about any constant stationary point ϕ_s. Each Fourier mode is then an oscillator, as in §15.1, with

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + U''(\phi_s). \tag{20.3}$$

At a minimum, $U'' > 0$, every $\omega_{\mathbf{k}}$ is real and the configuration is stable [hw. p. 4]. At a maximum, $U'' < 0$, the modes with $\mathbf{k}^2 < |U''|$ have imaginary frequency. They are

inverted oscillators and grow like $e^{\sqrt{|U''|-\mathbf{k}^2}\,t}$, the longest wavelengths fastest. This *tachyonic instability* is how a field rolls off a maximum: it drives spontaneous symmetry breaking and the end of inflation. A local minimum that is not the global one is a *false vacuum*, stable at every order of perturbation theory but not beyond (§20.6).

The constant ϕ_0 characterizes the vacuum, the state of lowest energy of the canonically quantized theory [hw. p. 4, comment 4], at the classical level. Quantum fluctuations correct this picture (§20.6, Chapter 29).



(a) the potential (b) shifted to $\chi = \phi - \phi_0$

Figure 37: (a) A potential with a true vacuum ϕ_0, a false vacuum ϕ_1 and a maximum between them [hw. p. 5]. (b) The same potential about ϕ_0. The dashed parabola is the free part $\frac{1}{2}\mu^2\chi^2$, an oscillator for every mode; the difference between the two curves is the interaction.

20.3 Choosing a vacuum and shifting the field [hw. pp. 5–6]

Choose the vacuum to be studied, say ϕ_0 (Fig. 37). One may also choose a false vacuum: perturbation theory about it is consistent at every order, and its decay is a separate, non-perturbative question (§20.6). Measure the field from the chosen vacuum and the potential from its value there [hw. p. 5]:

$$\chi = \phi - \phi_0, \qquad V(\chi) = U(\chi + \phi_0) - U(\phi_0). \tag{20.4}$$

Since $U(\phi) = V(\chi) + U(\phi_0)$, the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{2}\,\partial_\mu\chi\,\partial^\mu\chi - V(\chi) - U(\phi_0). \tag{20.5}$$

Caution: two stray factors of $\frac{1}{2}$

The boxed Lagrangian on [hw. p. 5] reads $-\frac{1}{2}\partial\chi\cdot\partial\chi - \frac{1}{2}V(\chi) - \frac{1}{2}U(\phi_0)$. The potential terms carry no $\frac{1}{2}$: (20.5) follows from $U(\phi) = V(\chi) + U(\phi_0)$ alone.

The measure is unchanged, $\mathcal{D}\phi = \mathcal{D}\chi$, because shifting every integration variable by the same constant has unit Jacobian. The constant $-U(\phi_0)$ multiplies $e^{iS/\hbar}$ by the phase $e^{-iU(\phi_0)\,VT/\hbar}$, where VT is the volume of spacetime. It is a vacuum energy density $U(\phi_0)$, and it cancels in every normalized Green's function (§23.3). One may therefore drop it and work with

$$\mathcal{L}_{\phi_0} = -\frac{1}{2}\,\partial_\mu\chi\,\partial^\mu\chi - V(\chi), \qquad S_{\phi_0} = \int \mathrm{d}^D x\,\mathcal{L}_{\phi_0} \tag{20.6}$$

[hw. pp. 5–6]. Gravity is the exception, since it responds to the vacuum energy itself (§24.4). In terms of χ,

$$\langle t_f, [\chi_f]|\mathcal{T}[\cdots]|t_i, [\chi_i]\rangle = \mathcal{N}_{\phi_0}\int \mathcal{D}\chi\; e^{iS_{\phi_0}[\chi]/\hbar}\,[\cdots], \qquad \chi_{i,f}(\mathbf{x}) = \phi_{i,f}(\mathbf{x}) - \phi_0 \tag{20.7}$$

[hw. p. 6].

20.4 Free part and interactions [hw. pp. 6–7]

Expand V about its minimum [hw. p. 6]. By construction $V(0) = 0$ and $V'(0) = 0$, so

$$\begin{gathered} V(\chi) = \frac{1}{2}\mu^2\chi^2 + \frac{g_3}{3!}\chi^3 + \frac{g_4}{4!}\chi^4 + \cdots, \\ \mu^2 = U''(\phi_0), \qquad g_3 = U'''(\phi_0), \qquad g_4 = U''''(\phi_0). \end{gathered} \tag{20.8}$$

The handwritten notes write m^2 and λ_3, λ_4. With $\hbar$ explicit the curvature of the potential is an inverse length squared: it is $\mu^2 = m^2/\hbar^2$, the field-side mass parameter of Part II (eq. (13.11)). The mass of the particle is the curvature of the potential at the vacuum.

- The quadratic term is the free Klein–Gordon field of §15. Its field equation is linear, every mode is an independent oscillator, and it is exactly solvable [hw. p. 6].
- The cubic and higher terms make the field equation nonlinear. They are the interactions, and they are not exactly solvable in general. Their strength is set by $g_3, g_4, \ldots$ [hw. p. 6].

Part II's diagrams already had this structure: the free part supplies the lines, the interactions the vertices [hw. p. 6, "structure of WL theory is emerging"]. Chapter 25 makes it precise.

The computation therefore faces two difficulties [hw. p. 6]: the nonlinearity, controlled by the couplings, and the non-classical paths, controlled by $\hbar$. Neither can be

handled exactly, and the remedy is an expansion in both, perturbative quantum field theory [hw. p. 7]. With this understood, rename χ back to ϕ [hw. p. 7]:

$$\begin{aligned} U(\phi) &= \frac{1}{2}\mu^2\phi^2 + \frac{g_3}{3!}\phi^3 + \frac{g_4}{4!}\phi^4 + \cdots , \\ \mathcal{L} &= -\frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\mu^2\phi^2 - \frac{g_3}{3!}\phi^3 - \frac{g_4}{4!}\phi^4 - \cdots . \end{aligned} \tag{20.9}$$

> **Caution: the sign of the interaction terms**
>
> The Lagrangian on [hw. p. 15] has $+\frac{\lambda_3}{3!}\phi^3 + \frac{\lambda_4}{4!}\phi^4$. The signs must be negative, as in (20.9): $\mathcal{L}$ contains $-U$, which is how p. 7 and p. 20 write it. With a plus sign the potential would contain $-\frac{g_4}{4!}\phi^4$, unbounded below for $g_4 > 0$, and the theory would have no vacuum at all.

20.5 Counting $\hbar$: dimensions and the loop expansion★

Dimensions. With $c = 1$ and $\hbar$ explicit, the action has the dimension of $\hbar$. The kinetic term $\int \mathrm{d}^D x\,(\partial\phi)^2$ then fixes the dimension of the field, and the other terms fix those of the parameters:

$$[\phi] = \hbar^{1/2}\,L^{1-D/2}, \qquad [\mu] = L^{-1}, \qquad [g_n] = \hbar^{1-n/2}\,L^{\,n(D-2)/2-D}. \tag{20.10}$$

One expansion, not two. Rescale $\phi = \sqrt{\hbar}\,\varphi$. Then

$$\frac{S}{\hbar} = \int \mathrm{d}^D x \Big[-\frac{1}{2}(\partial\varphi)^2 - \frac{1}{2}\mu^2\varphi^2 - \sum_{n\geq 3}\frac{\hat{g}_n}{n!}\varphi^n \Big], \qquad \hat{g}_n \equiv \hbar^{\,n/2-1} g_n. \tag{20.11}$$

Planck's constant survives only in the combinations $\hat{g}_3 = \sqrt{\hbar}\,g_3$ and $\hat{g}_4 = \hbar g_4$. At fixed g_n, expanding in $\hbar$ is the same as expanding in the effective couplings $\hat{g}_n$, so the "power series in both the λ's and $\hbar$" of [hw. p. 7] is a single expansion. By (20.10), $\hat{g}_n$ has the dimension $L^{n(D-2)/2-D}$. It is a pure number for ϕ^4 in $D = 4$ and for ϕ^3 in $D = 6$; in $D = 4$, $\hbar g_4$ is dimensionless and $\sqrt{\hbar}\,g_3$ is an inverse length. These dimensions decide which theories can be renormalized (§30.2).

Classical plus quantum. Any result can be written as a classical part plus quantum corrections [hw. p. 7]. The classical part is the leading order in $\hbar$: the tree diagrams, which still contain every order in the couplings because the field equation is nonlinear. §28.4 shows that the tree diagrams are exactly the iterative solution of the classical field equation. Each closed loop costs one more power of $\hbar$: a connected diagram with E external points and L loops is of order $\hbar^{E-1+L}$ (§17.8, derived in §26.5). The loop expansion is the stationary-phase expansion of §4.6 for fields. It is asymptotic rather than convergent (§21.5).

20.6 Vacuum structure beyond perturbation theory★

The vacuum picture of this chapter is exact at the classical level but receives quantum corrections [hw. p. 7]. They are of three kinds.

Perturbative shifts. Fluctuations about ϕ_0 shift the minimum. The quantum vacuum is the minimum of the effective potential (Chapter 29), not of U. With $g_3 \neq 0$ the shift appears at order $\hbar$, as a tadpole (§29.4), so the identification of the vacuum with $\phi = 0$ [hw. p. 15] holds at tree level only.

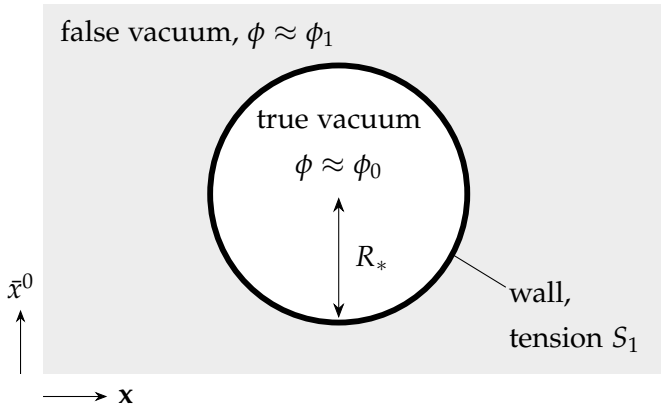


Figure 38: The thin-wall bounce: a ball of true vacuum in Euclidean spacetime. Making it larger gains the energy density ε inside and costs wall tension; the balance at $R_* = 3S_1/\varepsilon$ sets the decay rate $e^{-B/\hbar}$ of the false vacuum.

Degenerate vacua: quantum mechanics versus field theory. Let U have two equal minima at $\pm v$. In quantum mechanics a particle in a symmetric double well tunnels between them; the ground state is unique and symmetric, and the tunnelling splits the two lowest levels by an amount of order $e^{-S_{\text{inst}}/\hbar}$ (the instantons of §8.8). In field theory the whole field must tunnel. In Euclidean time the configuration that goes from $\phi = -v$ everywhere to $\phi = +v$ everywhere contains a domain wall stretched across all of space, with action σV_N, where σ is the wall's action per unit area and V_N the volume of space. The amplitude $e^{-\sigma V_N/\hbar}$ vanishes in infinite volume. The two vacua never mix: each is the ground state of its own sector, and the symmetry $\phi \to -\phi$ of U is spontaneously broken. This can only happen with infinitely many degrees of freedom.

False vacua decay. A false vacuum ϕ_1 (Fig. 37) lies above the true one by an energy density ε. It decays by nucleating bubbles of true vacuum, like a superheated liquid boiling. The rate is controlled by a Euclidean solution, the *bounce*. When ε is small, the

bounce is a ball of true vacuum of radius R in four-dimensional Euclidean spacetime, with a thin wall of tension S_1 (Fig. 38). Its action is the cost of the wall minus the gain of the interior:

$$B(R) = 2\pi^2 R^3 S_1 - \frac{\pi^2}{2} R^4 \varepsilon, \qquad \frac{\mathrm{d}B}{\mathrm{d}R} = 0 \implies R_* = \frac{3S_1}{\varepsilon}, \quad B = \frac{27\pi^2 S_1^4}{2\varepsilon^3}, \tag{20.12}$$

using the area $2\pi^2 R^3$ of a three-sphere and the volume $\frac{\pi^2}{2} R^4$ of a four-ball. The decay rate per unit volume is $\Gamma/V = A\, e^{-B/\hbar}$ (Coleman, 1977). Since B grows as the couplings shrink, $e^{-B/\hbar}$ has a vanishing Taylor series: false-vacuum decay is invisible at every order of perturbation theory, like the instantons mentioned in §17.8.

The cubic theory has no stable vacuum. With g_3 alone, $U = \frac{1}{2}\mu^2\phi^2 + \frac{g_3}{3!}\phi^3$ is unbounded below, so the vacuum at $\phi = 0$ is a false vacuum. Perturbation theory in g_3 is still meaningful, as the expansion about that false vacuum, and ϕ^3 is a convenient toy model for diagrams (§17.8). A term $g_4 > 0$ makes the theory stable.

21 Warm-up I: a field theory in zero dimensions★

Before the general machinery, here is the same machinery on a spacetime that is a single point. The "field" is one real number x and the "path integral" is an ordinary integral. Every combinatorial statement of Chapters 24–27 (Wick's theorem, Feynman diagrams, symmetry factors, the cancellation of vacuum bubbles, the exponentiation of connected pieces) can be checked here by expanding a function in a Taylor series. The Euclidean form is used, because its weight is real. The real-time version has extra factors of i but exactly the same combinatorics.

21.1 The integral and its Gaussian part

Take

$$Z(J) = \frac{\int \mathrm{d}x\; e^{-S(x)/\hbar + Jx}}{\int \mathrm{d}x\; e^{-\mu^2 x^2/2\hbar}}, \qquad S(x) = \frac{1}{2}\mu^2 x^2 + \frac{g_3}{3!}x^3 + \frac{g_4}{4!}x^4. \tag{21.1}$$

Derivatives with respect to the source bring down powers of x, so the averages $\langle x^n \rangle = Z^{(n)}(0)/Z(0)$ play the role of the N-point functions. Without interactions, completing the square gives

$$Z_0(J) = e^{\frac{1}{2}\Delta J^2}, \qquad \Delta = \frac{\hbar}{\mu^2}, \qquad \langle x^{2n} \rangle_0 = (2n-1)!!\;\Delta^n, \quad \langle x^{2n+1} \rangle_0 = 0. \tag{21.2}$$

The "propagator" Δ is the inverse of the quadratic form, divided by $\hbar$. The number $(2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n-1)$ counts the ways of splitting $2n$ objects into pairs: this is Wick's theorem in its simplest form. For example $\langle x^4 \rangle_0 = 3\Delta^2$, one term for each of the three pairings of Fig. 33.

21.2 Diagrams by Taylor expansion

With interactions, expand $e^{-(g_3 x^3/3! + g_4 x^4/4!)/\hbar}$ in powers of the couplings. Each term is a Gaussian moment, a sum over pairings, and each pairing can be drawn: every factor of x is an end, every vertex $x^n/n!$ is a dot with n legs, and every pair is a line. Collecting identical pictures gives the Feynman rules:

Feynman rules in zero dimensions
Each line gives Δ; each vertex with n legs gives $-g_n/\hbar$; each diagram is divided by its *symmetry factor* S, the number of ways of relabelling its lines and vertices that leave it unchanged.

The factor $1/S$ is what survives of the $1/v!$ from the exponential and the $1/n!$ of each vertex after counting the pairings that produce the same picture. Here is a check

of each rule against the integral. Write $\lambda_3 \equiv g_3^2\hbar/\mu^6$ and $\lambda_4 \equiv g_4\hbar/\mu^4$; these are the dimensionless combinations of §20.5.

- **Vacuum, quartic, first order.** $-\frac{g_4}{4!\hbar}\langle x^4\rangle_0 = -\frac{g_4}{4!\hbar}\,3\Delta^2 = -\frac{1}{8}\lambda_4$. This is the figure-eight, Fig. 39(a), with $S = 8$: each loop can be reversed, and the two loops can be exchanged.
- **Vacuum, cubic, second order.** $\frac{1}{2}\big(\frac{g_3}{3!\hbar}\big)^2\langle x^6\rangle_0 = \frac{15}{72}\lambda_3 = \frac{5}{24}\lambda_3$. Two diagrams contribute: the theta, Fig. 39(d), with $S = 3!\cdot 2 = 12$, and the dumbbell, Fig. 39(e), with $S = 8$. Indeed $\frac{1}{12} + \frac{1}{8} = \frac{5}{24}$.
- **One-point function, cubic.** $\langle x\rangle = -\frac{1}{2}\frac{g_3}{\hbar}\Delta^2 = -\frac{g_3\hbar}{2\mu^4}$: the tadpole, Fig. 39(f), with $S = 2$. The interaction pushes the average away from zero.
- **Two-point function, cubic, second order.** The integral gives $\langle x^2\rangle = \Delta\big(1 + \frac{5}{4}\lambda_3\big)$. The diagrams are the bubble (g), $S = 2$; the tadpole attached to the line (h), $S = 2$; and two separate tadpoles (i), $S = 4$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{4} = \frac{5}{4}$.
- **Quartic, first order.** $\langle x^2\rangle = \Delta\big(1 - \frac{1}{2}\lambda_4\big)$, the tadpole on the line (j) with $S = 2$; and the four-point function has the connected part $\langle x^4\rangle - 3\langle x^2\rangle^2 = -\frac{g_4}{\hbar}\Delta^4$, the bare vertex (k) with $S = 1$.

These symmetry factors are topological. The same diagrams with the same factors appear in D dimensions (§26.4); only the value of a line changes, $\Delta \to G_{\rm F}$, and each vertex is integrated over spacetime.

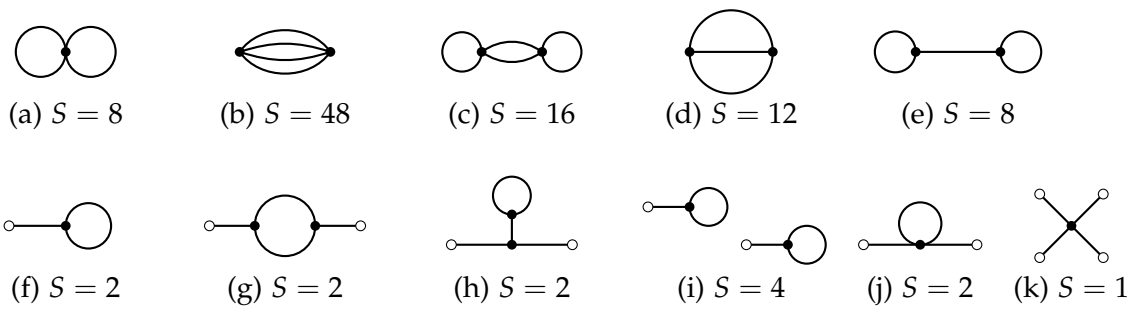


Figure 39: Diagrams and their symmetry factors. Filled dots are vertices, open dots external points. Top row, vacuum diagrams: (a)–(c) quartic, (d)–(e) cubic. Bottom row: (f) the cubic tadpole; (g)–(i) the cubic two-point function at second order, where (i) is disconnected; (j) the quartic two-point function at first order; (k) the quartic four-point vertex. The same numbers hold in any dimension.

21.3 Vacuum bubbles cancel, connected pieces exponentiate

Dividing by $Z(0)$ removes every diagram that contains a vacuum bubble, a piece with no external point: in $\langle x^2\rangle$ above, no such diagrams appear. The logarithm of Z does

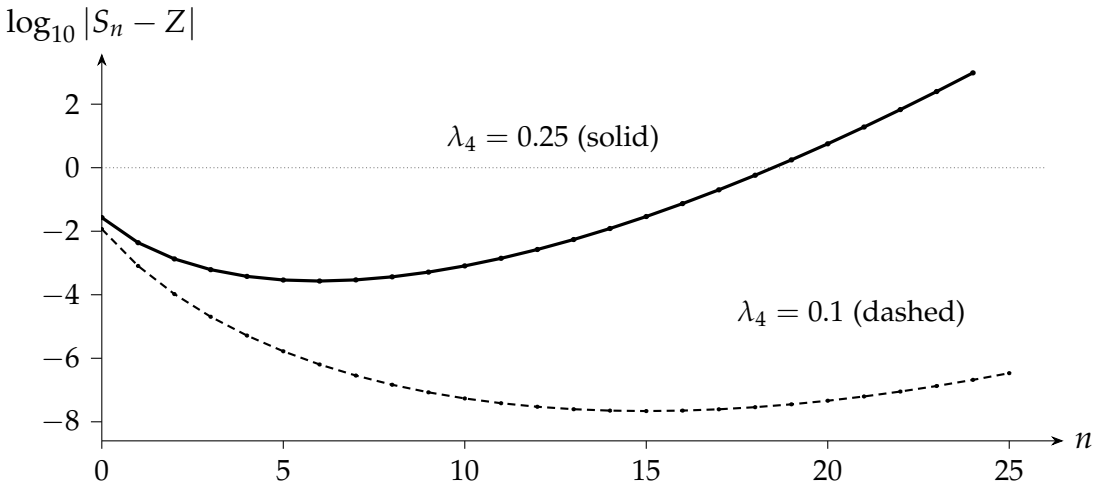


Figure 40: Error of the partial sums $S_n = \sum_{k\le n} c_k \lambda_4^k$ of the quartic integral. The error falls until $n_* \approx 3/(2\lambda_4)$ and then grows without bound: the series is asymptotic.

more. For the quartic theory

$$Z(0) = 1 - \frac{1}{8}\lambda_4 + \frac{35}{384}\lambda_4^2 + \cdots, \qquad \ln Z(0) = -\frac{1}{8}\lambda_4 + \frac{1}{12}\lambda_4^2 + \cdots. \tag{21.3}$$

At second order $Z(0)$ contains three diagrams: the two connected ones of Fig. 39(b) and (c), with $\frac{1}{48} + \frac{1}{16} = \frac{1}{12}$, and two separate figure-eights, with $\frac{1}{2!}\left(\frac{1}{8}\right)^2 = \frac{1}{128}$, the $\frac{1}{2!}$ counting the exchange of the identical pieces. In total $\frac{1}{12} + \frac{1}{128} = \frac{35}{384}$. The logarithm keeps only the connected diagrams. This is the exponentiation theorem of Chapter 27 in its simplest instance.

21.4 Connected functions are cumulants

$W(J) \equiv \ln Z(J)$ generates the *cumulants* of the distribution $e^{-S/\hbar}$:

$$\begin{aligned} \langle x^2\rangle_{\mathrm{c}} &= \langle x^2\rangle - \langle x\rangle^2, \qquad \langle x^3\rangle_{\mathrm{c}} = \langle x^3\rangle - 3\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3, \\ \langle x^4\rangle_{\mathrm{c}} &= \langle x^4\rangle - 4\langle x^3\rangle\langle x\rangle - 3\langle x^2\rangle^2 + 12\langle x^2\rangle\langle x\rangle^2 - 6\langle x\rangle^4, \end{aligned} \tag{21.4}$$

with $\langle x^n\rangle_{\mathrm{c}} = W^{(n)}(0)$. For the Gaussian, $W_0 = \frac{1}{2}\Delta J^2$ is quadratic, so every cumulant beyond the second vanishes: that is Wick's theorem in one line. In the cubic example, $\langle x^2\rangle_{\mathrm{c}} = \Delta(1 + \lambda_3)$, the bubble plus the tadpole on the line; the product of two tadpoles, Fig. 39(i), is $\langle x\rangle^2$ and drops out. Connected diagrams are cumulants. In probability theory the cumulants of independent random variables add; here the independent pieces of a diagram multiply in Z and add in $W = \ln Z$.

21.5 The perturbation series is asymptotic

For the quartic theory the whole series is known. With $\langle x^{4n}\rangle_0 = (4n-1)!!\,\Delta^{2n}$,

$$Z(0) = \sum_{n\geq 0} c_n \lambda_4^n, \qquad c_n = \frac{(-1)^n\,(4n-1)!!}{24^n\,n!}, \qquad \left|\frac{c_{n+1}}{c_n}\right| = \frac{(4n+1)(4n+3)}{24(n+1)} \approx \frac{2n}{3}. \tag{21.5}$$

The coefficients grow factorially and the radius of convergence is zero. Dyson's argument explains why without computing anything: for $\lambda_4 < 0$ the integral diverges, so Z cannot be analytic at $\lambda_4 = 0$. The series is *asymptotic*. The partial sums approach the exact value until the terms are smallest, near $n_* \approx 3/(2\lambda_4)$, with an accuracy of order $e^{-3/(2\lambda_4)}$ up to a prefactor, and then run away (Fig. 40). For $\lambda_4 = 0.25$ the best order is 6, with error 3×10^{-4}; for $\lambda_4 = 0.1$ it is 15, with error 2×10^{-8}. The accuracy that cannot be reached, $e^{-\text{const}/\lambda_4}$, is the size of the non-perturbative effects of §20.6. In field theory the number of diagrams at order n also grows like $n!$, and the loop expansion is asymptotic in the same way.

22 Warm-up II: a driven oscillator is a field theory in one dimension★

The second warm-up has the same structure as a free field theory, but its spacetime is the time axis alone. Its "field" is the coordinate $x(t)$ of a harmonic oscillator of unit mass, the system of §9 and the building block of every free field (§15.1). A source $J(t)$ pushes on it. Everything can be computed exactly, and every result has a direct physical meaning.

22.1 The vacuum-to-vacuum amplitude

Let the source act during a finite time and couple as $\hbar J(t)\,x(t)$, the normalization of [hw. p. 12]:

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2 + \hbar J(t)\,x. \tag{22.1}$$

Start in the ground state, let the source act, and ask for the amplitude to find the ground state at the end:

$$Z_0[J] \equiv \langle 0,\text{out}|0,\text{in}\rangle_J = \frac{\int \mathcal{D}x\; e^{iS_J[x]/\hbar}}{\int \mathcal{D}x\; e^{iS_0[x]/\hbar}}, \tag{22.2}$$

where both path integrals run over a long time with the projection onto the ground state of eq. (8.7). Chapter 23 shows that this projection is the replacement $\omega^2 \to \omega^2 - i\eta$, so take it as given here. In frequency space, $x(t) = \int \frac{\mathrm{d}E}{2\pi}\,\tilde{x}(E)e^{-iEt}$ and $\tilde{J}(E) = \int \mathrm{d}t\, J(t)\,e^{iEt}$, the action is

$$S_J = \int \frac{\mathrm{d}E}{2\pi}\Big[\frac{1}{2}\big(E^2 - \omega^2 + i\eta\big)\tilde{x}(E)\tilde{x}(-E) + \hbar\tilde{J}(-E)\tilde{x}(E)\Big]. \tag{22.3}$$

Shift $\tilde{x}(E) = \tilde{x}'(E) - \hbar\tilde{J}(E)/(E^2 - \omega^2 + i\eta)$. The terms linear in $\tilde{x}'$ cancel, the $\tilde{x}'$-integral cancels against the denominator, and what is left is

$$Z_0[J] = \exp\Big\{-\frac{i\hbar}{2}\int \frac{\mathrm{d}E}{2\pi}\,\frac{\tilde{J}(E)\tilde{J}(-E)}{E^2 - \omega^2 + i\eta}\Big\}. \tag{22.4}$$

Transforming back with eq. (15.7):

The driven oscillator

$$Z_0[J] = \exp\Big\{-\frac{1}{2}\int \mathrm{d}t\,\mathrm{d}t'\; J(t)\,G_{\mathrm{F}}(t-t')\,J(t')\Big\},$$
$$G_{\mathrm{F}}(t) = \langle 0|\mathcal{T}\,\hat{x}(t)\hat{x}(0)|0\rangle = \frac{\hbar}{2\omega}\,e^{-i\omega|t|}. \tag{22.5}$$

The amplitude is the exponential of one line, the oscillator's Feynman function, joining the source to itself. The $i\eta$ placed the poles of $1/(E^2-\omega^2+i\eta)$ as in Fig. 29, and that placement is what made the answer $G_{\rm F}$ rather than the retarded function.

22.2 A static source shifts the energy

Let the source rise slowly to a constant value J_0, stay there for a long time T, and fall slowly again. For $T \gg 1/\omega$ the double integral is $T\int \mathrm{d}\tau\, G_{\rm F}(\tau)$, and $\int \mathrm{d}\tau\, e^{-i\omega|\tau|} = 2/(i\omega)$, so

$$\ln Z_0 = -\frac{1}{2}J_0^2\, T\,\frac{\hbar}{2\omega}\,\frac{2}{i\omega} = \frac{i\hbar J_0^2}{2\omega^2}\,T \equiv -\frac{i}{\hbar}\,\Delta E\, T, \qquad \Delta E = -\frac{\hbar^2 J_0^2}{2\omega^2}. \tag{22.6}$$

The amplitude is a pure phase, and the phase is the energy shift of the ground state in the presence of the source. It equals the classical result: the potential $\frac{1}{2}\omega^2x^2 - \hbar J_0 x$ has its minimum at $x = \hbar J_0/\omega^2$, with value $-\hbar^2 J_0^2/2\omega^2$. For a linear system the source only displaces the oscillator, so the quantum and classical shifts agree exactly. The same calculation for a field gives the Yukawa potential (§24.3).

22.3 The source emits quanta, with Poisson statistics

A source that changes quickly leaves the oscillator excited. The probability that it stays in the ground state is

$$|Z_0[J]|^2 = \exp\Big\{-\frac{\hbar}{2\omega}\int \mathrm{d}t\,\mathrm{d}t'\, J(t)J(t')\cos\omega(t-t')\Big\} = e^{-\bar n}, \qquad \bar n = \frac{\hbar}{2\omega}\,\big|\tilde J(\omega)\big|^2. \tag{22.7}$$

The operator solution says what happens instead. The Heisenberg equation is $\ddot{\hat x} + \omega^2\hat x = \hbar J$, and once the source is off its retarded solution gives

$$\hat a_{\rm out} = \hat a_{\rm in} + \alpha, \qquad \alpha = i\sqrt{\frac{\hbar}{2\omega}}\;\tilde J(\omega). \tag{22.8}$$

The initial ground state satisfies $\hat a_{\rm out}|0,\mathrm{in}\rangle = \alpha|0,\mathrm{in}\rangle$: it is a coherent state of the final oscillator. The number of quanta found at the end is Poisson distributed,

$$P(n) = e^{-\bar n}\,\frac{\bar n^{\,n}}{n!}, \qquad \bar n = |\alpha|^2, \tag{22.9}$$

and $P(0)$ agrees with (22.7). Only the Fourier component of the source at the oscillator's frequency matters. A source switched on and off slowly has $\tilde J(\omega) \approx 0$ and emits nothing; the ground state follows it adiabatically, which is why the static source of §22.2 produced a pure phase.

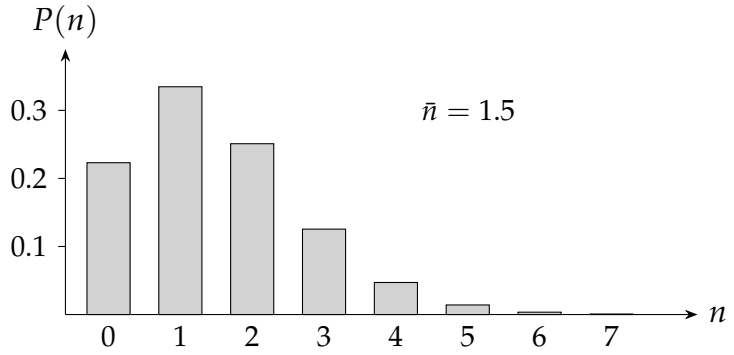


Figure 41: The number of quanta left behind by a source, here with $\bar{n} = 1.5$: a Poisson distribution. The probability of none is $e^{-\bar{n}} = |Z_0[J]|^2$.

22.4 Diagrams: why the exponential

$W_0 \equiv \ln Z_0 = -\frac{1}{2}\int\!\!\int J\, G_{\rm F}\, J$ is a single line whose two ends sit on the source: the "dumbbell" of [hw. p. 17]. Its exponential is a sum over any number of disjoint dumbbells,

$$Z_0 = \sum_{n\geq 0} \frac{W_0^n}{n!}, \tag{22.10}$$

each standing for one quantum emitted and reabsorbed by the source. The quanta are independent, so their amplitudes multiply, and they are identical, which gives the $1/n!$. The Poisson distribution (22.9) carries the same $1/n!$ for the same reason: it is the statistics of independent, identical events, and the $1/n!$ is the Gibbs factor for identical particles. This is the physical content of the exponentiation theorem of Chapter 27: the full generating functional is the exponential of the connected one because connected pieces are independent.

22.5 One driven oscillator per mode

A free field is one oscillator of unit mass for each wave vector, with frequency $\omega_{\mathbf{k}}$ (§15.1). A source $J(t,\mathbf{x})$ drives mode $\mathbf{k}$ through its Fourier component $\tilde{J}(t,\mathbf{k}) = \int \frac{\mathrm{d}^N x}{(2\pi)^{N/2}} e^{-i\mathbf{k}\cdot\mathbf{x}} J(t,\mathbf{x})$, because $\hbar\int \mathrm{d}^D x\, J\phi = \hbar\int \mathrm{d}t \int \mathrm{d}^N k\, \tilde{J}(t,-\mathbf{k})\tilde{\phi}(t,\mathbf{k})$. The modes do not talk to each other, so the amplitude is the product of single-oscillator results (22.5):

$$\begin{aligned} Z_0[J] &= \exp\Big\{ -\frac{1}{2}\int \mathrm{d}^N k \int \mathrm{d}t\, \mathrm{d}t'\; \tilde{J}(t,-\mathbf{k})\, \frac{\hbar}{2\omega_{\mathbf{k}}} e^{-i\omega_{\mathbf{k}}|t-t'|}\; \tilde{J}(t',\mathbf{k}) \Big\} \\ &= \exp\Big\{ -\frac{1}{2}\int \mathrm{d}^D x\, \mathrm{d}^D y\; J(x)\, G_{\rm F}(x-y)\, J(y) \Big\}, \end{aligned} \tag{22.11}$$

with $G_{\rm F}(x-y) = \int \frac{\mathrm{d}^N k}{(2\pi)^N} \frac{\hbar}{2\omega_{\mathbf{k}}}\, e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y}) - i\omega_{\mathbf{k}}|x^0-y^0|}$, the field's Feynman function (Appendix C of Part II). At the level of the free field, nothing beyond this chapter is needed; Chapter 24 derives the same result covariantly from the path integral.

23 Green's functions from the path integral

23.1 Green's functions with boundary conditions [hw. p. 8]

Define the N-point function "with boundary conditions" [hw. p. 8]:

$$\begin{aligned} G_{if}^{(N)}(x_1,\ldots,x_N) &\equiv \langle t_f,[\phi_f]|\mathcal{T}\big[\hat\phi(x_1)\cdots\hat\phi(x_N)\big]|t_i,[\phi_i]\rangle \\ &= \mathcal{N}\int_{\phi_i}^{\phi_f}\mathcal{D}\phi\; e^{iS[\phi]/\hbar}\,\phi(x_1)\cdots\phi(x_N), \end{aligned} \tag{23.1}$$

with every x_k^0 between t_i and t_f. How is it related to the diagrammatic expressions of Part II? For $N=2$ in the free theory, the line of Part II is $G_{\rm F}(x_1-x_2) = \langle 0|\mathcal{T}\,\hat\phi(x_1)\hat\phi(x_2)|0\rangle$, a vacuum expectation value. A plausible guess [hw. p. 8] is therefore that the diagrams compute

$$G^{(N)}(x_1,\ldots,x_N) = \langle\Omega|\mathcal{T}\big[\hat\phi(x_1)\cdots\hat\phi(x_N)\big]|\Omega\rangle, \tag{23.2}$$

where $|\Omega\rangle$ is the vacuum of the interacting theory. Two questions follow [hw. p. 8]: what is the path-integral representation of (23.2), and which diagrams does it produce?

23.2 Projecting onto the vacuum [hw. pp. 9–10]

Write the states as $|t,[\phi]\rangle = e^{i\hat H t/\hbar}|[\phi]\rangle$ and insert complete sets of energy eigenstates $|m\rangle$ on both sides of the time-ordered product [hw. p. 9]. The set contains $|\Omega\rangle$ and the excited states, which in the free theory are $|\mathbf{k}\rangle$, $|\mathbf{k}_1,\mathbf{k}_2\rangle$, … Then

$$G_{if}^{(N)} = \sum_{m,n}\Psi_m[\phi_f]\;e^{-iE_m t_f/\hbar}\;\langle m|\mathcal{T}[\cdots]|n\rangle\;e^{iE_n t_i/\hbar}\;\Psi_n^*[\phi_i], \qquad \Psi_m[\phi]=\langle[\phi]|m\rangle, \tag{23.3}$$

where Ψ_m are the wave functionals of §18.4. All dependence on t_i and t_f sits in the exponentials, so the endpoints can be moved into the complex plane. Take them along a slightly rotated line, $t_f = T(1-i\eta)$ and $t_i = -T(1-i\eta)$ with $T\to\infty$. Then $e^{-iE_m t_f/\hbar} = e^{-iE_m T/\hbar}\,e^{-\eta E_m T/\hbar}$, and relative to the vacuum term every excited state is suppressed by $e^{-\eta(E_m-E_\Omega)T/\hbar}\to 0$ [hw. pp. 9–10]. (The handwritten notes rotate all the way, $t\to -iu$ with $u_f\to+\infty$ and $u_i\to-\infty$; the infinitesimal rotation is enough.) What survives is

$$G_{if}^{(N)} \;\longrightarrow\; \Psi_\Omega[\phi_f]\,\Psi_\Omega^*[\phi_i]\;e^{-iE_\Omega(t_f-t_i)/\hbar}\;\langle\Omega|\mathcal{T}[\hat\phi(x_1)\cdots\hat\phi(x_N)]|\Omega\rangle. \tag{23.4}$$

Caution: what the projection needs

- The handwritten notes set $E_\Omega = 0$ and assume "a bounded system" [hw. p. 10]. Neither is needed, and E_Ω is generally not zero: it is the vacuum energy, infinite in the continuum (§24.4). What is needed is a spectrum bounded below, with the excited states separated from E_Ω by a gap; in a massive theory the gap is the energy $\hbar\mu$ of one particle at rest. This is the same correction as in Part I (footnote in §8.3). Without a gap, in a massless theory, the suppression is only a power of T.
- The limit does not exist in strictly real time, where the excited states oscillate instead of dying out. The tilt is essential.
- The overlaps $\Psi_\Omega[\phi_f]$ and $\Psi_\Omega[\phi_i]$ must not vanish. They never do: the ground state of a bosonic system has no nodes, like the oscillator ground state, and the free vacuum functional (18.8) is a Gaussian, positive everywhere. (On [hw. p. 11] the second factor lacks its complex conjugate.)

23.3 The ratio formula [hw. p. 11]

Without insertions, the same argument gives the denominator [hw. p. 11]:

$$\langle t_f, [\phi_f] | t_i, [\phi_i] \rangle \;\longrightarrow\; \Psi_\Omega[\phi_f]\, \Psi^*_\Omega[\phi_i]\; e^{-iE_\Omega(t_f - t_i)/\hbar}. \tag{23.5}$$

In the ratio the wave functionals and the vacuum energy cancel:

Vacuum Green's functions from the path integral

$$G^{(N)}(x_1, \ldots, x_N) = \langle \Omega | \mathcal{T}\big[\hat\phi(x_1) \cdots \hat\phi(x_N)\big] | \Omega \rangle = \frac{\displaystyle\int \mathcal{D}\phi\; e^{iS[\phi]/\hbar}\, \phi(x_1) \cdots \phi(x_N)}{\displaystyle\int \mathcal{D}\phi\; e^{iS[\phi]/\hbar}}, \tag{23.6}$$

with $S = \int \mathrm{d}t \int \mathrm{d}^N x\, \mathcal{L}$ and the time integral running from $-\infty(1 - i\eta)$ to $+\infty(1 - i\eta)$.

Numerator and denominator both depend on the boundary configurations, but the ratio does not [hw. p. 12]. Equation (23.6) is Part I's eq. (8.7), the Gell-Mann–Low formula, for a field. It answers the first question of §23.1; the diagrams it produces are the subject of Chapters 24–27.

23.4 The vacuum projection is the $i\eta$★

The tilted time contour can be traded for a term in the action. Let $t = \tau(1 - i\eta)$ with real τ. Then $\mathrm{d}t = (1 - i\eta)\mathrm{d}\tau$ and $\partial_t = (1 - i\eta)^{-1}\partial_\tau$, and to first order in η the free action becomes

$$S_0 = \int \mathrm{d}\tau\, \mathrm{d}^N x \left[\frac{1}{2}(1 + i\eta)(\partial_\tau\phi)^2 - \frac{1}{2}(1 - i\eta)\left((\nabla\phi)^2 + \mu^2\phi^2\right)\right]. \tag{23.7}$$

For a plane wave the bracket is $-\frac{1}{2}\left[k^2 + \mu^2 - i\eta\,(k_0^2 + \omega_{\mathbf{k}}^2)\right]\tilde{\phi}\tilde{\phi}$. The coefficient of $i\eta$ is positive, and only its sign matters, so:

The vacuum projection is $\mu^2 \to \mu^2 - i\eta$
Taking the time contour to $\pm\infty(1 - i\eta)$ is the same as replacing $\mu^2 \to \mu^2 - i\eta$ in the action. Equivalently $e^{iS/\hbar}$ acquires the factor $e^{-(\eta/2\hbar)\int \mathrm{d}^D x\, \phi^2}$, a Gaussian damping that makes the integral converge, and propagators acquire the Feynman placement of poles.

It is the same infinitesimal clockwise rotation, met for the third time:

where	what is rotated	prescription
Part I, §8.9	time t	$m\omega^2 \to m\omega^2 - i\eta$
Part II, §13.4	proper-time length T	$m^2 \to m^2 - i\eta$
here	field time x^0	$\mu^2 \to \mu^2 - i\eta$

The boundary behaviour that the handwritten notes leave implicit ("Fourier transform that automatically assumes the boundary behaviour", [hw. p. 14]) is this $i\eta$. It selects the vacuum at both ends, and therefore the Feynman function among all Green's functions of the Klein–Gordon operator (§15.7).

23.5 Euclidean field theory★

Rotating all the way, $x^0 = -i\bar{x}^0$, turns iS into $-S_{\mathrm{E}}$ with the Euclidean action

$$S_{\mathrm{E}}[\phi] = \int \mathrm{d}^D\bar{x}\left[\frac{1}{2}\left(\bar{\partial}\phi\right)^2 + U(\phi)\right], \tag{23.8}$$

which is bounded below whenever U is. The Euclidean Green's functions

$$G_{\mathrm{E}}^{(N)}(\bar{x}_1, \ldots, \bar{x}_N) = \frac{\int \mathcal{D}\phi\; e^{-S_{\mathrm{E}}[\phi]/\hbar}\,\phi(\bar{x}_1)\cdots\phi(\bar{x}_N)}{\int \mathcal{D}\phi\; e^{-S_{\mathrm{E}}[\phi]/\hbar}} \tag{23.9}$$

are averages with a positive weight, and on a lattice they are ordinary convergent integrals. For the free field, $G_{\mathrm{E}}(\bar{x} - \bar{y}) = \hbar\int \frac{\mathrm{d}^D\bar{k}}{(2\pi)^D}\,\frac{e^{i\bar{k}\cdot(\bar{x}-\bar{y})}}{\bar{k}^2+\mu^2}$, the function $-i\bar{\Delta}$ of eq. (15.9).

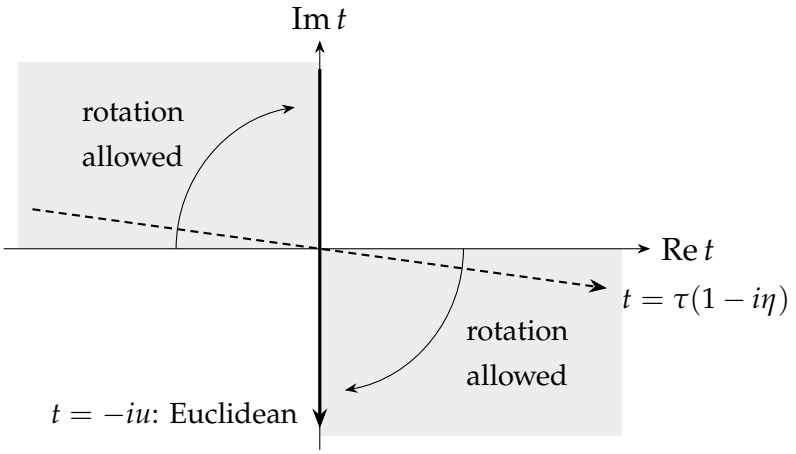


Figure 42: The time contour of the field path integral. The vacuum projection tilts it infinitesimally clockwise (dashed); the Wick rotation turns it all the way onto the imaginary axis. The rotation sweeps only the shaded quadrants, where the integrand decays. Compare Fig. 14 and Fig. 26.

The real-time functions are recovered by continuing back, $\bar{x}^0 \to ix^0$; the conditions under which this works are the Osterwalder–Schrader axioms, which the Euclidean functions of a sensible theory satisfy.

Three readings carry over from Part I.

- **Statistical mechanics.** $e^{-S_E/\hbar}$ is a Boltzmann weight for a classical field in D dimensions, with $\hbar$ in the role of the temperature (§8.7). Quantum field theory in N space dimensions is classical statistical mechanics in $D = N + 1$ dimensions; $1/\mu$ is the correlation length, and a massless field is a system at its critical point.
- **Zero temperature.** The vacuum projection of §23.2 is the limit of an infinitely long Euclidean time, $\hbar\beta \to \infty$: zero temperature (§8.11). Keeping the Euclidean time periodic with period $\hbar\beta$ instead gives field theory at temperature $1/\beta$, through the partition function (8.20).
- **Convergence.** With U bounded below every integral converges absolutely, and every manipulation of the following chapters (shifting variables, exchanging integrals, integrating by parts) is justified there and carried to real time by the $i\eta$.

23.6 The generating functional [hw. pp. 12–14]

Computing (23.6) for every N at once is the job of the generating functional [hw. p. 12]. Add a source to the action, $S[\phi, J] = S[\phi] + \hbar(J, \phi)$ with $(J, \phi) = \int \mathrm{d}^D x\, J(x)\phi(x)$ [hw. p. 12], where J is a fixed external scalar field, and define

$$\mathcal{Z}[J] = \int \mathcal{D}\phi\ \exp\left\{\frac{i}{\hbar}S[\phi] + i(J,\phi)\right\}, \qquad Z[J] = \frac{\mathcal{Z}[J]}{\mathcal{Z}[0]} \tag{23.10}$$

(the handwritten $W[J]$). The boundary conditions are implicit: they are the vacuum ones, through the $i\eta$, and the source vanishes at $t \to \pm\infty$. Expanding in the source [hw. p. 12; solution of the exercise in Appendix E.1],

$$Z[J] = \sum_{N\geq 0} \frac{1}{N!} \int \mathrm{d}^D x_1 \cdots \mathrm{d}^D x_N \; Z^{(N)}(x_1, \ldots, x_N)\, J(x_1) \cdots J(x_N),$$
$$Z^{(N)} = \frac{\delta^N Z}{\delta J(x_1) \cdots \delta J(x_N)}\Big|_{J=0}. \tag{23.11}$$

Each functional derivative brings down one field [hw. p. 13]:

$$\frac{\delta}{\delta J(x_1)}\, e^{i(J,\phi)} = i \int \mathrm{d}^D y\, \delta^D(y - x_1)\, \phi(y)\, e^{i(J,\phi)} = i\phi(x_1)\, e^{i(J,\phi)}. \tag{23.12}$$

Hence $Z^{(N)} = i^N G^{(N)}$ [hw. p. 13], and:

Green's functions from the generating functional

$$G^{(N)}(x_1, \ldots, x_N) = \langle \Omega | \mathcal{T}[\hat\phi(x_1) \cdots \hat\phi(x_N)] | \Omega \rangle = \frac{\delta}{i\,\delta J(x_1)} \cdots \frac{\delta}{i\,\delta J(x_N)}\, Z[J]\Big|_{J=0} \tag{23.13}$$

[hw. p. 14]. Physically, $Z[J] = \langle \Omega, \mathrm{out} | \Omega, \mathrm{in} \rangle_J$ is the amplitude for the vacuum to survive the source, as for the oscillator of Chapter 22.

The recap of [hw. p. 14] is now a claim with content: the N-point functions that define the theory, computed in Part II by worldline diagrammatics, are vacuum expectation values of time-ordered products, and they are derivatives of $Z[J]$. It remains to compute $Z[J]$, first without and then with interactions.

24 The free field

24.1 The free generating functional [hw. pp. 15–17]

Take the free theory, $S_0[\phi] = -\frac{1}{2}\int \mathrm{d}^D x\,(\partial_\mu\phi\,\partial^\mu\phi + \mu^2\phi^2)$, and transform the field and the source in spacetime [hw. p. 15]: $\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^{D/2}}\, e^{ik\cdot x}\,\tilde\phi(k)$, and the same for J. With $\int \frac{\mathrm{d}^D x}{(2\pi)^D}\, e^{i(k+q)\cdot x} = \delta^D(k+q)$ [hw. p. 16], the action with source becomes [hw. p. 15; Appendix E.2]

$$S_0[\phi] + \hbar(J,\phi) = \int \mathrm{d}^D k\,\Big[-\frac{1}{2}\big(k^2+\mu^2-i\eta\big)\tilde\phi(k)\tilde\phi(-k) + \hbar\tilde J(k)\tilde\phi(-k)\Big], \tag{24.1}$$

with the $i\eta$ of §23.4. (The handwritten line has $\phi(-p)$ for $\tilde\phi(-k)$.) Complete the square by the constant shift [hw. p. 16; Appendix E.3]

$$\tilde\phi(k) = \tilde\phi'(k) + \frac{\hbar\,\tilde J(k)}{k^2+\mu^2-i\eta}, \tag{24.2}$$

which gives

$$S_0 + \hbar(J,\phi) = \int \mathrm{d}^D k\,\Big[-\frac{1}{2}\big(k^2+\mu^2-i\eta\big)\tilde\phi'(k)\tilde\phi'(-k) + \frac{\hbar^2}{2}\,\frac{\tilde J(k)\tilde J(-k)}{k^2+\mu^2-i\eta}\Big]. \tag{24.3}$$

The shift has unit Jacobian, $\mathcal{D}\phi' = \mathcal{D}\phi$ [hw. p. 16], so the ϕ'-integral is $\mathcal{Z}_0[0]$ and cancels in $Z_0[J] = \mathcal{Z}_0[J]/\mathcal{Z}_0[0]$. Transforming back to position space [hw. p. 17; Appendix E.4]:

The free generating functional

$$Z_0[J] = \exp\Big\{\frac{i}{2}(J,\Delta J)\Big\} = \exp\Big\{-\frac{1}{2}(J,G_{\mathrm{F}}J)\Big\},$$
$$\Delta(x-y) = \hbar\int \frac{\mathrm{d}^D k}{(2\pi)^D}\,\frac{e^{ik\cdot(x-y)}}{k^2+\mu^2-i\eta}, \tag{24.4}$$

where $(J,\Delta J) \equiv \int \mathrm{d}^D x\,\mathrm{d}^D y\, J(x)\Delta(x-y)J(y)$ and $G_{\mathrm{F}} = -i\Delta$.

This Δ is Part II's eq. (15.8), so the path integral reproduces the Feynman function of canonical quantization, and (24.4) agrees with the product of driven oscillators, eq. (22.11).

> **Caution: the $i\eta$ is not optional**
>
> The handwritten notes say that Δ "should be interpreted with the $i\epsilon$ prescription or Euclideanization" [hw. p. 17]. Without it the Gaussian integral over ϕ' does not converge and $1/(k^2+\mu^2)$ has poles on the real k^0-axis, so neither the shift nor the answer is defined. With it, the answer is the Feynman function and not, say, the retarded one: the vacuum boundary conditions of §23.4 are built in.

24.2 Wick's theorem and the free diagrams [hw. pp. 17–19]

Expand the exponential and sort the terms by the number n of sources [hw. p. 17]: $Z_0[J] = \sum_n T_0^{(n)}$. Odd n do not occur. Writing each factor of the exponent as $\frac{1}{2}(iJ, G_{\mathrm{F}}\, iJ)$, a line G_{F} with a source iJ at each end,

$$T_0^{(0)} = \langle 0|0\rangle = 1, \qquad T_0^{(2)} = \frac{1}{2}(iJ, G_{\mathrm{F}}\, iJ), \qquad T_0^{(2n)} = \frac{1}{n!}\Big[\frac{1}{2}(iJ, G_{\mathrm{F}}\, iJ)\Big]^n \tag{24.5}$$

[hw. pp. 17–18; on p. 17 the first factor of $T_0^{(4)}$ should have iJ_x, not $-iJ_x$]. The term $T_0^{(2n)}$ is n disjoint "dumbbells" (Fig. 43), and its symmetry factor $1/S = 1/(n!\,2^n)$ counts the exchange of the two ends of each dumbbell and the permutations of the identical dumbbells [hw. p. 18]. The Feynman rules for Z_0 are [hw. p. 18]: a source dot gives $i\int \mathrm{d}^D x\, J(x)$; a line gives $G_{\mathrm{F}}(x-y) = \frac{1}{i}\Delta(x-y)$; divide by S.

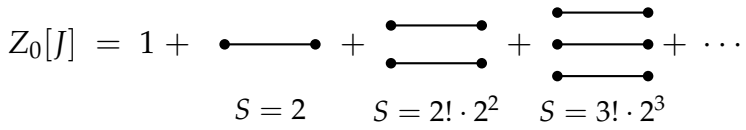


Figure 43: The free generating functional as a sum of disjoint dumbbells [hw. p. 18]. Each dot is a factor $i\int J$, each line a factor G_{F}; n identical dumbbells carry $1/(n!\,2^n)$.

The Green's functions follow by differentiation [hw. pp. 18–19]. For $N = 2$ the two derivatives can act on the two ends of the dumbbell in two ways, which cancels the $\frac{1}{2}$:

$$G^{(2)}(x_1, x_2) = \frac{\delta}{i\,\delta J(x_1)}\frac{\delta}{i\,\delta J(x_2)}\, T_0^{(2)} = G_{\mathrm{F}}(x_1 - x_2). \tag{24.6}$$

For $N = 4$ the first derivative may take any dot; the dot at the other end of its dumbbell then has three possible partners, and the remaining pair is fixed [hw. p. 19]:

$$G^{(4)} = G_{\mathrm{F}}(x_1-x_2)G_{\mathrm{F}}(x_3-x_4) + G_{\mathrm{F}}(x_1-x_3)G_{\mathrm{F}}(x_2-x_4) + G_{\mathrm{F}}(x_1-x_4)G_{\mathrm{F}}(x_2-x_3), \tag{24.7}$$

Part II's eq. (17.9), now derived. In general [hw. p. 19]: the free $G^{(N)}$ is the sum over all pairings of the N external points, with a factor G_{F} for each pair and symmetry factor 1, because the external points are labelled. This is Wick's theorem, derived from the generating functional as §17.6 promised. Its content is that Z_0 is a Gaussian, whose connected part $W_0 = \ln Z_0 = -\frac{1}{2}(J, G_{\mathrm{F}}J)$ is quadratic: free particles never scatter (§27.3).

24.3 A first physical result: the Yukawa potential★

Part II claimed that the exchange of a massive particle produces a short-range force (§13.7). Here is the derivation. Take two static point sources in $N = 3$, $J(x) =$

$j_1\delta^3(\mathbf{x}-\mathbf{x}_1)+j_2\delta^3(\mathbf{x}-\mathbf{x}_2)$, switched on slowly for a long time T, as for the oscillator of §22.2. The cross term of the exponent of (24.4) is

$$\frac{i}{2}(J,\Delta J)\supset ij_1j_2\int \mathrm{d}t\int \mathrm{d}t'\,\Delta(t-t',\mathbf{r}),\qquad \mathbf{r}=\mathbf{x}_1-\mathbf{x}_2. \tag{24.8}$$

The integral over the time difference sets $k^0=0$: $\int \mathrm{d}t'\,\Delta(t-t',\mathbf{r})=\hbar\int\frac{\mathrm{d}^3k}{(2\pi)^3}\frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\mathbf{k}^2+\mu^2}=\hbar\,\frac{e^{-\mu r}}{4\pi r}$. Writing $Z_0=e^{-iET/\hbar}$,

The Yukawa potential
$$E(r)=-\hbar^2 j_1 j_2\,\frac{e^{-\mu r}}{4\pi r}. \tag{24.9}$$

The force has range $1/\mu=\hbar/mc$, the Compton wavelength of the exchanged particle; Yukawa (1935) used this to predict the pion mass from the range of the nuclear force. Sources of the same sign *attract*: exchanging a spin-0 particle is attractive. A spin-1 particle, the photon, makes like charges repel, and spin 2, gravity, always attracts. As $\mu\to 0$ the energy becomes a $1/r$ law. The terms in j_1^2 and j_2^2 do not depend on r. They are the self-energies of the point sources, and they are infinite, $e^{-\mu r}/r$ at $r=0$: this is the first ultraviolet divergence in these notes. It does not affect the force.

24.4 The vacuum functional and the zero-point energy★

Without sources, $\mathcal{Z}_0[0]=\int\mathcal{D}\phi\,e^{iS_0/\hbar}$ is the vacuum-to-vacuum amplitude of the free field over the time T, $\langle 0|e^{-i\hat{H}_0T/\hbar}|0\rangle=e^{-iE_0T/\hbar}$. The Gaussian integral gives a determinant,

$$\mathcal{Z}_0[0]\propto\det{}^{-1/2}\big(-\partial^2+\mu^2-i\eta\big)=\exp\Big\{-\frac{1}{2}\,\mathrm{Tr}\ln\big(-\partial^2+\mu^2-i\eta\big)\Big\}, \tag{24.10}$$

and in momentum space $\mathrm{Tr}\ln = VT\int\frac{\mathrm{d}^Dk}{(2\pi)^D}\ln(k^2+\mu^2-i\eta)$, with VT the volume of spacetime. The k^0-integral is $\int\frac{\mathrm{d}k^0}{2\pi}\ln\big(-k_0^2+\omega_{\mathbf{k}}^2-i\eta\big)=i\omega_{\mathbf{k}}+\text{const}$ (Appendix E.5), so

$$\ln\mathcal{Z}_0[0]=-\frac{i}{2}\,VT\int\frac{\mathrm{d}^Nk}{(2\pi)^N}\,\omega_{\mathbf{k}}\equiv-\frac{i}{\hbar}\,E_0T,\qquad E_0=V\int\frac{\mathrm{d}^Nk}{(2\pi)^N}\,\frac{1}{2}\hbar\omega_{\mathbf{k}}. \tag{24.11}$$

The path integral knows about the zero-point energy $\frac{1}{2}\hbar\omega_{\mathbf{k}}$ of every mode, the infinite sum that Part II left for this part (§15.1). In proper-time form it is the closed worldline of eq. (17.10): in Euclidean time, $\ln\mathcal{Z}_{0,\mathrm{E}}[0]=-\frac{1}{2}\,\mathrm{Tr}\ln(-\bar{\partial}^2+\mu^2)=-E_0T_\mathrm{E}/\hbar$. For a massless field in $D=4$ with a cutoff Λ on $|\mathbf{k}|$, $E_0/V=\hbar\Lambda^4/16\pi^2$, which diverges as $\Lambda\to\infty$. Three facts make this harmless or interesting, depending on what one asks.

1. **It cancels in every Green's function.** E_0 drops out of the ratio (23.6); in operator language, $\hat{H}$ is normal-ordered (§15.1). Without gravity only energy differences are observable.

2. **Differences are physical: the Casimir effect.** Put a massless scalar in $1+1$ dimensions between two plates a distance a apart, with $\phi = 0$ on the plates. The modes are $\omega_n = n\pi/a$ (Fig. 44). With a smooth cutoff,

$$\begin{aligned} E(a) &= \frac{\hbar}{2} \sum_{n \geq 1} \omega_n \, e^{-\omega_n/\Lambda} \\ &= \frac{\hbar \Lambda^2}{2\pi} \, a - \frac{\pi \hbar}{24 \, a} + O\big(\Lambda^{-2}\big) \end{aligned} \tag{24.12}$$

   (Appendix E.6). The first term is the energy density of empty space,

$$\int \frac{\mathrm{d}k}{2\pi} \frac{1}{2} \hbar |k| \, e^{-|k|/\Lambda} = \frac{\hbar \Lambda^2}{2\pi}, \tag{24.13}$$

   times the length a. It is there with or without the plates, and the space outside them holds the rest, so it produces no force. The second term does not depend on the cutoff. It gives an attractive force $F = -\pi\hbar c/24a^2$. Between parallel plates in $3+1$ dimensions a scalar gives $E/A = -\pi^2 \hbar c/1440\, a^3$; the electromagnetic field, with two polarizations, gives twice that, and the force has been measured.

3. **Gravity sees the absolute value.** In general relativity the vacuum energy density acts as a cosmological constant. With the cutoff at the Planck scale the estimate exceeds the observed value by roughly 120 orders of magnitude: the cosmological constant problem, the most visible place where quantum field theory and gravity collide.

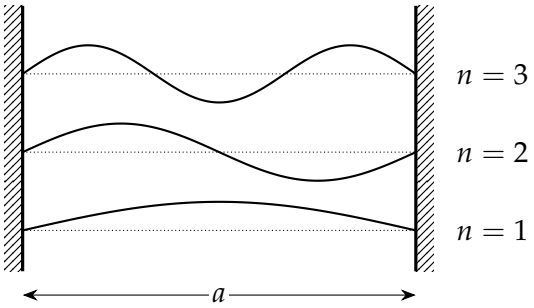


Figure 44: The lowest modes of a field held at zero on two plates. Their zero-point energies depend on the separation a, and the cutoff-independent part, $-\pi\hbar c/24a$ in $1+1$ dimensions, pulls the plates together.

25 The free field as a gas of worldlines★

Part II computed one worldline at a time: the amplitude for one particle to go from y to x. The field computes all of them at once. This chapter makes the handwritten remark that "the structure of the worldline theory is emerging" [hw. p. 6] precise. It is stated in Euclidean signature, where every weight is positive and the pictures can be taken literally.

25.1 Lines are worldlines

With the normalization $\alpha = 4/\hbar^2$, the worldline path integral of Part II equals the Feynman function, eq. (16.8). In Euclidean form, eq. (13.14) with $4/(\alpha\hbar) = \hbar$,

$$G_{\mathrm{E}}(\bar{x}-\bar{y}) = \hbar \int_0^\infty \mathrm{d}\sigma\, (4\pi\sigma)^{-D/2}\, e^{-|\bar{x}-\bar{y}|^2/4\sigma}\, e^{-\mu^2\sigma}. \tag{25.1}$$

This is a sum over Brownian paths from $\bar{y}$ to $\bar{x}$ of every duration σ, each weighted by $e^{-\mu^2\sigma}$ (§13.7): a line of a Feynman diagram is a worldline summed over its shapes and lengths (§17.1). The Euclidean free generating functional is

$$Z_{0,\mathrm{E}}[J] = \frac{\int \mathcal{D}\phi\; e^{-S_{0,\mathrm{E}}/\hbar + (J,\phi)}}{\int \mathcal{D}\phi\; e^{-S_{0,\mathrm{E}}/\hbar}} = \exp\Big\{\frac{1}{2}(J, G_{\mathrm{E}} J)\Big\}, \tag{25.2}$$

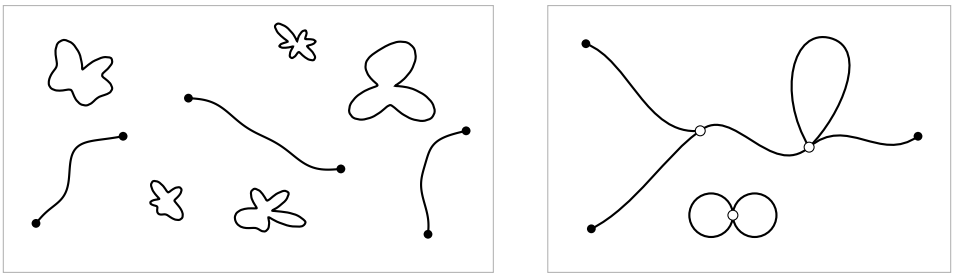
and its exponent is one open worldline with both ends on the source, with a factor $\frac{1}{2}$ because the two ends are interchangeable.

25.2 Loops are closed worldlines

Without sources, the unnormalized functional is the closed worldline of eq. (17.10):

$$\ln \mathcal{Z}_{0,\mathrm{E}}[0] = -\frac{1}{2}\,\mathrm{Tr}\ln\big(-\bar{\partial}^2 + \mu^2\big) = \frac{1}{2}\int_0^\infty \frac{\mathrm{d}\sigma}{\sigma} \int \mathrm{d}^D\bar{x}\; (4\pi\sigma)^{-D/2}\, e^{-\mu^2\sigma} + \text{const.} \tag{25.3}$$

That is one closed worldline, of any length and anywhere in spacetime. The $1/\sigma$ undoes the marked starting point and the $\frac{1}{2}$ identifies the two orientations (§17.7, Fig. 34).



(a) free: loops and strands (b) interacting: strands meet

Figure 45: (a) The free field as an ideal gas: closed worldlines (vacuum loops) and open worldlines whose ends sit on the source (filled dots), all independent. (b) With interactions, worldlines meet at vertices (open circles): a cubic vertex splits a strand, a quartic vertex joins four.

25.3 The ideal gas of worldlines

The free field is an ideal gas of worldlines

Let $\mathcal{L}_\circ$ be the sum over one closed worldline (25.3) and $\mathcal{O} = \frac{1}{2}(J, G_{\rm E} J)$ the sum over one open worldline ending on the source (25.2). Then

$$\mathcal{Z}_{0,{\rm E}}[J] = e^{\mathcal{L}_\circ + \mathcal{O}} = \sum_{n,m\geq 0} \frac{\mathcal{L}_\circ^n}{n!}\,\frac{\mathcal{O}^m}{m!}\,: \tag{25.4}$$

any number of closed loops and open strands, which do not interact, with Gibbs factors $1/n!$ and $1/m!$ for identical objects.

This is the grand-canonical partition function of an ideal gas whose particles are worldlines (Fig. 45a), Symanzik's random-path representation of the free field (1969). First and second quantization are two ways of organizing the same sum. Part II computed one member of the gas; the field path integral sums the whole gas with the right combinatorics, including the closed loops, the vacuum fluctuations, for which a one-particle theory has no room. The statement of §17.6 that free particles are independent is the statement that the gas is ideal.

25.4 Interactions: worldlines that meet

Interactions let worldlines meet (Fig. 45b). A cubic vertex is a point where one worldline splits into two; a quartic vertex is a point where four strands touch. In Euclidean ϕ^4 theory with $g_4 > 0$ the factor $e^{-(g_4/4!\hbar)\int \phi^4}$ penalizes configurations in which strands pass through the same point: it is a repulsive contact interaction between worldlines, and expanding it generates the vertices of Chapter 26. The two ingredients of a theory named in Part II, a propagator and a topology (§17.9), are exactly this: worldlines, and

the rule for how they meet.

Intuition: polymers are worldlines that avoid themselves

A long flexible polymer in solution is a random walk that cannot cross itself. Edwards modelled it as a Brownian path with a repulsive contact interaction, which is the worldline of ϕ^4 theory. De Gennes (1972) made the connection exact: self-avoiding walks are described by an $O(n)$-symmetric ϕ^4 theory in the formal limit $n \to 0$, which removes the closed loops and leaves a single strand. The renormalization group of Chapter 30 then predicts how the size of the polymer grows with its length, $R \propto N^{\nu}$ with $\nu \approx 0.588$ in three dimensions instead of the free random walk's $\frac{1}{2}$. Part I's polymer (Fig. 17) is the free version of the same picture.

26 The interacting theory: Feynman rules derived

26.1 The interacting generating functional [hw. pp. 20–21]

With interactions, $S[\phi, J] = \int \mathrm{d}^D x \left[\mathcal{L}_0 - V_{\text{int}}(\phi) + \hbar J\phi\right]$ [hw. p. 20], where $\mathcal{L}_0$ is the free Klein–Gordon Lagrangian and $V_{\text{int}} = \frac{g_3}{3!}\phi^3 + \frac{g_4}{4!}\phi^4 + \cdots$. The Green's functions are still (23.6) and (23.13), but the interaction sits in the numerator and in the denominator of $Z[J]$, which is awkward [hw. p. 20]. Normalize by the free vacuum instead [hw. p. 21]:

$$\tilde{Z}[J] \equiv \frac{\mathcal{Z}[J]}{\mathcal{Z}_0[0]} = \frac{\int \mathcal{D}\phi\; e^{\frac{i}{\hbar}S[\phi,J]}}{\int \mathcal{D}\phi\; e^{\frac{i}{\hbar}S_0[\phi]}}, \qquad \tilde{Z}[J] = \tilde{Z}[0]\, Z[J], \qquad Z[0] = \langle\Omega|\Omega\rangle = 1. \tag{26.1}$$

The key step [hw. p. 21] uses $\frac{\delta}{i\,\delta J(x)}\, e^{i(J,\phi)} = \phi(x)\, e^{i(J,\phi)}$: inside the integral every field can be replaced by a derivative with respect to the source, so for any functional F,

$$\int \mathcal{D}\phi\; F[\phi]\; e^{\frac{i}{\hbar}S_0[\phi]+i(J,\phi)} = F\Big[\frac{\delta}{i\,\delta J}\Big] \int \mathcal{D}\phi\; e^{\frac{i}{\hbar}S_0[\phi]+i(J,\phi)}. \tag{26.2}$$

Taking $F = \exp\{-\frac{i}{\hbar}\int \mathrm{d}^D y\, V_{\text{int}}(\phi(y))\}$:

The interacting generating functional

$$\tilde{Z}[J] = \exp\Big\{-\frac{i}{\hbar}\int \mathrm{d}^D y\; V_{\text{int}}\Big(\frac{\delta}{i\,\delta J(y)}\Big)\Big\}\; Z_0[J], \qquad Z_0[J] = \exp\Big\{-\frac{1}{2}(J, G_{\text{F}} J)\Big\} \tag{26.3}$$

[hw. pp. 21–22]. The interaction acts as a differential operator on the free Gaussian.

26.2 The double expansion [hw. p. 22]

Expanding both exponentials [hw. p. 22],

$$\tilde{Z}[J] = \sum_{v\geq 0} \frac{1}{v!}\Big[-\frac{i}{\hbar}\int \mathrm{d}^D y\; V_{\text{int}}\Big(\frac{\delta}{i\,\delta J(y)}\Big)\Big]^v \sum_{p\geq 0} \frac{1}{p!}\Big[\frac{1}{2}(iJ, G_{\text{F}}\, iJ)\Big]^p. \tag{26.4}$$

(The handwritten "expanding around $\phi = 0$" means expanding in the couplings and the sources.) In a cubic theory, v vertices act with $3v$ derivatives on p dumbbells, which carry $2p$ sources, and leave $E = 2p - 3v$ of them. Every surviving term is a picture with v vertices, p lines and E source dots, and $\tilde{Z}[J] = \sum_n \tilde{T}^{(n)}$ collects them by the number n of sources [hw. p. 22]. The terms with no sources, $\tilde{T}^{(0)} = \tilde{Z}[0]$, are the vacuum diagrams [hw. pp. 22–23], such as those of Fig. 39(a)–(e).

26.3 The Feynman rules

Differentiating $Z = \tilde{Z}/\tilde{Z}[0]$ removes every diagram with a vacuum bubble (Chapter 27) and turns the remaining source dots into the external points. The result is the algorithm

of Goal I:

Feynman rules for $G^{(N)}(x_1,\dots,x_N)$

1. Draw every diagram with the N external points and any number of internal vertices, leaving out diagrams with a piece that has no external point.
2. Each line from a to b gives $G_{\mathrm{F}}(a-b)=\frac{1}{i}\Delta(a-b)$.
3. Each vertex with n legs at y gives $-\frac{i}{\hbar}\,g_n\int \mathrm{d}^D y$.
4. Divide by the symmetry factor S: the number of relabellings of the internal lines and vertices that leave the diagram unchanged, with the external points held fixed.

In momentum space, with $\tilde{G}^{(N)}(k_1,\dots,k_N)=\int\prod_j \mathrm{d}^D x_j\, e^{-ik_j\cdot x_j}\,G^{(N)}$ and all k_j flowing in: each line with momentum k gives $\dfrac{-i\hbar}{k^2+\mu^2-i\eta}$; each vertex gives $-\frac{i}{\hbar}g_n\,(2\pi)^D\delta^D(\sum k_{\mathrm{in}})$; every internal momentum is integrated with $\int\frac{\mathrm{d}^D k}{(2\pi)^D}$; divide by S.

The vertex has no $1/n!$ because the n derivatives at y can be attached to the n line ends in $n!$ ways, cancelling the $1/n!$ in V_{int}. The cancellation fails only when a relabelling maps the diagram to itself, and that is what S counts. These are the rules Part II stated in §17.8; the momentum-space line is the Fourier transform of $G_{\mathrm{F}}=-i\Delta$ in (24.4).

26.4 Worked examples

(a) The tree-level three-point function, cubic theory. At first order in g_3 the terms of (26.4) with three sources have $v=1$ and $p=3$: $-\frac{i}{\hbar}\frac{g_3}{3!}\int\mathrm{d}^D y\,\big(\frac{\delta}{i\,\delta J(y)}\big)^3\frac{1}{3!}\big[\frac{1}{2}(iJ,G_{\mathrm{F}}\,iJ)\big]^3$. The three derivatives at y take one end of each dumbbell, in $3!\cdot 2^3$ ways, and the derivatives at x_1,x_2,x_3 take the three remaining ends, in $3!$ ways. The factors $\frac{1}{3!}\cdot\frac{1}{3!}\cdot\frac{1}{2^3}$ cancel exactly, so $S=1$ and

$$G_{\mathrm{c}}^{(3)}(x_1,x_2,x_3)=-\frac{ig_3}{\hbar}\int\mathrm{d}^D y\;G_{\mathrm{F}}(x_1-y)\,G_{\mathrm{F}}(x_2-y)\,G_{\mathrm{F}}(x_3-y),\tag{26.5}$$

Part II's eq. (17.12), now derived. The same terms also contain the configuration in which two of the derivatives at y take both ends of one dumbbell: it gives the disconnected products $G_{\mathrm{F}}(x_1-x_2)\,G^{(1)}(x_3)$ and permutations, with the tadpole $G^{(1)}$ of (b).

(b) The tadpole. In the cubic theory the one-point function does not vanish at first order (Fig. 39f, $S = 2$):

$$G^{(1)}(x) = \langle\Omega|\hat{\phi}(x)|\Omega\rangle = \frac{1}{2}\Big(-\frac{ig_3}{\hbar}\Big)\int \mathrm{d}^D y\; G_{\mathrm{F}}(x-y)\, G_{\mathrm{F}}(0) = -\frac{g_3}{2\mu^2}\, G_{\mathrm{F}}(0), \quad (26.6)$$

using $\int \mathrm{d}^D y\, G_{\mathrm{F}}(x-y) = -i\hbar/\mu^2$, the momentum-space line at $k = 0$. Here $G_{\mathrm{F}}(0) = \langle 0|\hat{\phi}(x)^2|0\rangle > 0$ is the mean square fluctuation of the field, a divergent constant for $D \geq 2$ (§30.1). Classically the cubic term exerts the force $-\frac{g_3}{2}\phi^2$; averaged over the fluctuations, it displaces the field by exactly this amount (§28.3).

(c) The one-loop two-point function, cubic theory. At second order:

$$\begin{aligned} G^{(2)}(x_1, x_2) = {} & G_{\mathrm{F}}(x_1 - x_2) + \frac{1}{2}\Big(-\frac{ig_3}{\hbar}\Big)^2 \int \mathrm{d}^D y\, \mathrm{d}^D z\; G_{\mathrm{F}}(x_1 - y)\, G_{\mathrm{F}}(y-z)^2\, G_{\mathrm{F}}(z - x_2) \\ & + \frac{1}{2}\Big(-\frac{ig_3}{\hbar}\Big)^2 \int \mathrm{d}^D y\, \mathrm{d}^D z\; G_{\mathrm{F}}(x_1 - y)\, G_{\mathrm{F}}(y - x_2)\, G_{\mathrm{F}}(y-z)\, G_{\mathrm{F}}(0) \\ & + G^{(1)}(x_1)\, G^{(1)}(x_2). \end{aligned} \quad (26.7)$$

These are the bubble ($S = 2$, Part II's eq. (17.13)), the tadpole attached to the line ($S = 2$) and two separate tadpoles ($S = 4$): Fig. 39(g)–(i). The ordinary integral of §21.2 confirmed $\frac{1}{2} + \frac{1}{2} + \frac{1}{4}$.

(d) The quartic theory at first order. The two-point function acquires the tadpole on the line (Fig. 39j, $S = 2$), and the four-point function its first connected term ($S = 1$):

$$\begin{aligned} G^{(2)} &\supset \frac{1}{2}\Big(-\frac{ig_4}{\hbar}\Big)\int \mathrm{d}^D y\; G_{\mathrm{F}}(x_1 - y)\, G_{\mathrm{F}}(0)\, G_{\mathrm{F}}(y - x_2), \\ G_{\mathrm{c}}^{(4)} &= -\frac{ig_4}{\hbar}\int \mathrm{d}^D y \prod_{j=1}^{4} G_{\mathrm{F}}(x_j - y). \end{aligned} \quad (26.8)$$

In momentum space the first term is

$$\frac{1}{2}\Big(-\frac{ig_4}{\hbar}\Big) G_{\mathrm{F}}(0)\Big(\frac{-i\hbar}{k^2+\mu^2}\Big)^2 = \frac{i\hbar\,(g_4/2) G_{\mathrm{F}}(0)}{(k^2+\mu^2)^2}, \quad (26.9)$$

and comparing with $\frac{-i\hbar}{k^2+\mu^2+\delta\mu^2} = \frac{-i\hbar}{k^2+\mu^2} + \frac{i\hbar\,\delta\mu^2}{(k^2+\mu^2)^2} + \cdots$ shows that the tadpole shifts the mass:

$$\delta\mu^2 = \frac{g_4}{2}\, G_{\mathrm{F}}(0) = \frac{g_4}{2}\,\langle\phi^2\rangle. \quad (26.10)$$

This is the curvature of the potential, $U'' = \mu^2 + \frac{g_4}{2}\phi^2$, averaged over the fluctuations. It diverges, and it is the first loop divergence that renormalization will have to absorb (Chapter 30).

(e) The vacuum diagrams of [hw. p. 23]. The theta, the dumbbell and the figure-eight are

$$\frac{1}{12}\Big(-\frac{ig_3}{\hbar}\Big)^2\int \mathrm{d}^D y\,\mathrm{d}^D z\; G_{\mathrm{F}}(y-z)^3, \qquad \frac{1}{8}\Big(-\frac{ig_3}{\hbar}\Big)^2\int \mathrm{d}^D y\,\mathrm{d}^D z\; G_{\mathrm{F}}(0)\,G_{\mathrm{F}}(y-z)\,G_{\mathrm{F}}(0),$$
$$\frac{1}{8}\Big(-\frac{ig_4}{\hbar}\Big)\int \mathrm{d}^D y\; G_{\mathrm{F}}(0)^2, \tag{26.11}$$

with the symmetry factors 12, 8 and 8 found for the ordinary integral in §21.2. Each contains an unconstrained integral over the position of the whole diagram and is therefore proportional to the volume of spacetime (§27.4).

26.5 Counting $\hbar$

By the rules, every line carries one power of $\hbar$ (eq. (24.4)) and every vertex one power of $1/\hbar$. A connected diagram with E external points, I internal lines and v vertices has $E+I$ lines in all, so it is of order

$$\hbar^{E+I-v} = \hbar^{E-1+L}, \qquad L = I - v + 1, \tag{26.12}$$

where L is its number of loops. This is the counting stated in §17.8, now derived: tree diagrams are of order $\hbar^{E-1}$, and each loop costs one more power of $\hbar$. For example the tree (26.5) is of order $\hbar^{-1}\hbar^3 = \hbar^2$, the bubble in (26.7) of order $\hbar^{-2}\hbar^4 = \hbar^2$, and the tadpole (26.6) of order $\hbar$. The loop expansion is the expansion in the effective couplings $\hat{g}_n$ of §20.5.

27 Vacuum bubbles, connected diagrams and cumulants

The handwritten notes end with the structure of the sum over diagrams [hw. pp. 22–25]: the vacuum diagrams factor out, and connected diagrams exponentiate. The second statement implies the first, so it comes first here.

27.1 Connected pieces exponentiate [hw. pp. 24–25]

Three facts, already visible in the free theory (§24.2), hold in general [hw. p. 24]: a disconnected diagram is a product of connected pieces; every possible product appears in the generating functional; and n identical connected pieces carry an extra symmetry factor $1/n!$. The last one follows from the definition of S. An automorphism of a diagram either maps each connected piece to itself, or permutes identical pieces among themselves; it can never map a piece to a different one. So if a diagram contains n_I copies of the connected diagram C_I,

$$S(\text{diagram}) = \prod_I n_I!\; S(C_I)^{n_I}. \tag{27.1}$$

Let C_I now denote the *value* of the I-th connected diagram, its factor $1/S$ included. Summing over all diagrams means summing over all multiplicities $n_I \geq 0$ independently [hw. pp. 24–25]:

Exponentiation of connected diagrams

$$\sum_{\text{all diagrams}} = \sum_{\{n_I \geq 0\}} \prod_I \frac{C_I^{n_I}}{n_I!} = \prod_I e^{C_I} = \exp\Big(\sum_I C_I\Big). \tag{27.2}$$

Applied to $\tilde{Z}[J]$, with $C_I^{(0)}$ the connected vacuum diagrams and $C_I^{(N)}$ the connected diagrams with $N \geq 1$ source dots:

$$\tilde{Z}[J] = \exp\Big(\sum_{N\geq 0}\sum_I C_I^{(N)}\Big) = e^{\tilde{W}[0]}\, e^{W[J]}, \qquad \tilde{W}[0] = \sum_I C_I^{(0)}, \quad W[J] = \sum_{N\geq 1}\sum_I C_I^{(N)}. \tag{27.3}$$

In the handwritten notation $\tilde{W}[0]$ and $W[J]$ are $\tilde{Z}[0]$ and $Z[J]$ [hw. p. 25]. By construction $W[0] = 0$. For a given number of sources there are infinitely many connected diagrams, but only finitely many at each order in the couplings, so listing them is the well-defined problem of [hw. p. 24] order by order. The physical reason for (27.2) is the one found for the driven oscillator (§22.4): connected pieces are independent events, independent amplitudes multiply, and the $1/n_I!$ is the Gibbs factor for identical events.

27.2 Vacuum bubbles cancel [hw. pp. 22–23]

Comparing (27.3) with $\tilde{Z}[J] = \tilde{Z}[0]\, Z[J]$ from (26.1) gives at once

$$\tilde{Z}[0] = e^{\tilde{W}[0]}, \qquad Z[J] = e^{W[J]}. \tag{27.4}$$

The vacuum sector factors out completely. Order by order in the sources, $\tilde{T}^{(n)} = \tilde{T}^{(0)}\, T^{(n)}$, where $T^{(n)}$ contains no vacuum pieces [hw. p. 23]: dividing by $\tilde{Z}[0]$ removes every diagram that contains a bubble, which is why the rules of §26.3 leave them out. The handwritten notes state this before the exponentiation lemma [hw. p. 23]; it is a consequence of it. $\tilde{Z}[J]$ itself contains disconnected diagrams of every kind, with vacuum and non-vacuum pieces [hw. p. 23]; $Z[J] = e^{W[J]}$ still contains disconnected diagrams, but no vacuum pieces; and $W[J]$ is connected.

27.3 Connected Green's functions are cumulants

The connected Green's functions are the derivatives of $W = \ln Z$:

Connected Green's functions

$$G_{\mathrm{c}}^{(N)}(x_1, \ldots, x_N) = \frac{\delta}{i\,\delta J(x_1)} \cdots \frac{\delta}{i\,\delta J(x_N)}\, W[J]\Big|_{J=0}. \tag{27.5}$$

Since $Z = e^W$, every $G^{(N)}$ is a sum over the ways of splitting the N points into groups, with a factor G_{c} for each group:

$$\begin{aligned} G^{(1)} &= G_{\mathrm{c}}^{(1)}, \qquad G^{(2)}(1,2) = G_{\mathrm{c}}^{(2)}(1,2) + G_{\mathrm{c}}^{(1)}(1)\, G_{\mathrm{c}}^{(1)}(2), \\ G^{(4)}(1,2,3,4) &= G_{\mathrm{c}}^{(4)}(1,2,3,4) + \big[G_{\mathrm{c}}^{(2)}(1,2)\, G_{\mathrm{c}}^{(2)}(3,4) + 2 \text{ more}\big] \\ &\quad + \big(\text{terms with } G_{\mathrm{c}}^{(1)} \text{ or } G_{\mathrm{c}}^{(3)}\big). \end{aligned} \tag{27.6}$$

These are the relations between moments and cumulants of §21.4, for a probability distribution over field configurations. For the free field $W_0 = -\frac{1}{2}(J, G_{\mathrm{F}} J)$ is quadratic, so $G_{\mathrm{c}}^{(N)} = 0$ for $N \neq 2$: a Gaussian has no cumulants beyond the second. This is Wick's theorem in one line, and it says that free particles never scatter. In the cubic example (26.7), $G_{\mathrm{c}}^{(2)}$ is the free line plus the bubble plus the tadpole on the line; the product of two tadpoles is $G^{(1)}G^{(1)}$.

27.4 Why connected diagrams are extensive: the vacuum energy★

Every connected vacuum diagram contains one integral over the position of the whole diagram that nothing constrains, so it is proportional to the volume of spacetime, VT

(eq. (26.11)). On the other hand $\mathcal{Z}[0] \propto e^{-iE_\Omega T/\hbar}$ and $\mathcal{Z}_0[0] \propto e^{-iE_0 T/\hbar}$ (§23.2), so

$$\tilde{W}[0] = \ln \frac{\mathcal{Z}[0]}{\mathcal{Z}_0[0]} = -\frac{i}{\hbar}\,(E_\Omega - E_0)\,T, \qquad E_\Omega - E_0 = \frac{i\hbar}{T}\sum_I C_I^{(0)} \;\propto\; V. \tag{27.7}$$

The connected vacuum diagrams compute the shift of the vacuum energy caused by the interactions, and that shift is proportional to the volume of space. Exponentiation is what makes this possible: disconnected bubbles grow like $(VT)^2, (VT)^3, \ldots$, and they must assemble into an exponential for the energy to be extensive. It is the field-theory version of the statement that $\ln Z = -\beta F$ is extensive in statistical mechanics (§8.11).

Check: first-order perturbation theory. For the quartic theory the only connected vacuum diagram at first order is the figure-eight of (26.11), $C = \frac{1}{8}\big(-\frac{ig_4}{\hbar}\big)VT\,G_{\rm F}(0)^2$. Then (27.7) gives

$$E_\Omega - E_0 = \frac{g_4}{8}\,V\,\langle\phi^2\rangle^2, \tag{27.8}$$

which is first-order perturbation theory, $\langle 0|\int \mathrm{d}^N x\,\frac{g_4}{4!}\hat{\phi}^4|0\rangle = \frac{g_4}{24}\cdot 3\langle\phi^2\rangle^2\,V$, using $\langle\phi^4\rangle = 3\langle\phi^2\rangle^2$ from Wick's theorem. The figure-eight is the first-order energy shift of the vacuum.

28 The Schwinger–Dyson equations★

Feynman diagrams solve the theory order by order. The Schwinger–Dyson equations state it exactly, as the field equation imposed on the generating functional. They are the field version of Part I's eq. (7.16), and they rederive the results of the last three chapters in a few lines.

28.1 Integrating a total derivative

On a lattice the path integral is an ordinary multiple integral, and the integral of a derivative vanishes when the integrand decays at large field values; the Euclidean weight, or the damping of §23.4, guarantees that. In the continuum, for any functional F,

$$\int \mathcal{D}\phi \, \frac{\delta}{\delta\phi(x)} \Big\{ F[\phi] \, e^{\frac{i}{\hbar}S[\phi]+i(J,\phi)} \Big\} = 0. \tag{28.1}$$

At $J = 0$ this gives $\big\langle \frac{\delta S}{\delta\phi(x)} F \big\rangle = i\hbar \big\langle \frac{\delta F}{\delta\phi(x)} \big\rangle$, eq. (7.16) with $x(t) \to \phi(x)$. With $F = 1$ and the source kept, it gives $\big\langle \frac{\delta S}{\delta\phi(x)} \big\rangle_J = -\hbar J(x)$. Since $\frac{\delta S}{\delta\phi(x)} = -(-\partial^2 + \mu^2)\phi(x) - V'_{\text{int}}(\phi(x))$, and every field can be replaced by $\delta/i\delta J$:

Schwinger–Dyson equation for the generating functional

$$\Big[\big(-\partial_x^2 + \mu^2 \big) \frac{\delta}{i\,\delta J(x)} + V'_{\text{int}}\Big(\frac{\delta}{i\,\delta J(x)} \Big) \Big] \, Z[J] = \hbar J(x)\, Z[J]. \tag{28.2}$$

This is the classical field equation $(-\partial^2 + \mu^2)\phi + V'_{\text{int}}(\phi) = \hbar J$, imposed on $Z[J]$.

28.2 The free theory in one line

For $V_{\text{int}} = 0$ the equation is first order in $\delta/\delta J$. Try $Z_0 = \exp\{-\frac{1}{2}(J, GJ)\}$ with a symmetric kernel G. Then $\frac{\delta}{i\,\delta J(x)} Z_0 = i \int \mathrm{d}^D y \, G(x,y) J(y) \, Z_0$, and (28.2) holds for every J if and only if

$$\big(-\partial_x^2 + \mu^2 \big) \, G(x,y) = -i\hbar \, \delta^D(x-y). \tag{28.3}$$

So G is a Green's function of the Klein–Gordon operator, as Part II found from the worldline (eq. (14.6)) and from the field (eq. (15.6)). The equation does not say which one: the boundary conditions do, and the vacuum ones, the $i\eta$, select $G = G_{\text{F}}$. The completing of the square in §24.1 is replaced by one line.

28.3 The hierarchy and the contact terms

Differentiate (28.2) with respect to $J(y_1), \dots, J(y_{N-1})$ and set $J = 0$. On the right each derivative can hit the explicit $J(x)$ and produce a delta function:

$$\big(-\partial_x^2 + \mu^2\big)\big\langle \mathcal{T}\,\phi(x)\phi(y_1)\cdots\phi(y_{N-1})\big\rangle + \big\langle \mathcal{T}\,V'_{\text{int}}(\phi(x))\,\phi(y_1)\cdots\phi(y_{N-1})\big\rangle \\ = -i\hbar\sum_{j=1}^{N-1}\delta^D(x-y_j)\,\Big\langle \mathcal{T}\prod_{k\neq j}\phi(y_k)\Big\rangle. \tag{28.4}$$

The classical field equation holds inside time-ordered correlators, up to contact terms of order $\hbar$ at coincident points: Part I's statement after eq. (7.16), now for fields. The contact terms are those of eq. (7.11), and the seagull of §17.5 is the same phenomenon on the worldline.

Example: the tadpole in one line. Take $N = 1$ in the cubic theory. Then $(-\partial^2 + \mu^2)\langle\phi(x)\rangle + \frac{g_3}{2}\langle\phi(x)^2\rangle = 0$. The vacuum value is constant, and to lowest order $\langle\phi^2\rangle = G_{\text{F}}(0)$, so $\langle\phi\rangle = -\frac{g_3}{2\mu^2}\,G_{\text{F}}(0)$: the tadpole (26.6), without drawing a diagram.

Each equation of the hierarchy ties a Green's function to higher ones. The tower is exact and non-perturbative. Perturbation theory is its iteration, and its truncations (the Hartree approximation, the large-N expansion) are the standard approximations beyond perturbation theory.

28.4 Tree diagrams solve the classical field equation

Divide (28.2) by Z and write $\Phi_J(x) \equiv \langle\phi(x)\rangle_J = \frac{\delta}{i\,\delta J(x)}W[J]$ for the mean field in the presence of the source:

$$\big(-\partial^2 + \mu^2\big)\Phi_J + \langle V'_{\text{int}}(\phi)\rangle_J = \hbar J. \tag{28.5}$$

To leading order in $\hbar$ the fluctuations are negligible and $\langle V'_{\text{int}}(\phi)\rangle_J \approx V'_{\text{int}}(\Phi_J)$. The mean field then obeys the classical field equation with a source,

$$\big(-\partial^2 + \mu^2\big)\Phi + V'_{\text{int}}(\Phi) = \hbar J. \tag{28.6}$$

Solve it by iteration, using $(-\partial^2 + \mu^2)^{-1} = \frac{i}{\hbar}G_{\text{F}}$ from (28.3), with the vacuum boundary conditions. Writing $(G_{\text{F}} * f)(x) = \int \mathrm{d}^D y\, G_{\text{F}}(x-y)f(y)$,

$$\Phi = i\,G_{\text{F}} * J - \frac{i}{\hbar}\,G_{\text{F}} * V'_{\text{int}}(\Phi): \qquad \Phi_1 = i\,G_{\text{F}} * J, \qquad \Phi_2 = -\frac{i}{\hbar}\,\frac{g_3}{2}\,G_{\text{F}} * \Phi_1^2, \quad \dots \tag{28.7}$$

in the cubic theory. Φ_1 is one line ending on the source. Φ_2 is a vertex whose two branches end on the source, with the factor $\frac{1}{2}$ for the two identical branches, exactly

as the Feynman rules give it. Each iteration grafts new branches onto the old ones (Fig. 46): the solution of the classical field equation is the sum of all tree diagrams with one external point. This makes the "classical" part of [hw. p. 7] exact. The connected generating functional at tree level is then $W_{\text{tree}} = \frac{i}{\hbar}\big(S[\Phi_J] + \hbar(J, \Phi_J)\big)$, the stationary-phase value of the path integral (Chapter 29).

One feature is unfamiliar: because G_{F} carries the vacuum boundary conditions at both ends of time, this "classical" field is complex. It does not describe the classical evolution of an initial state; it describes the vacuum-to-vacuum amplitude in the presence of the source.

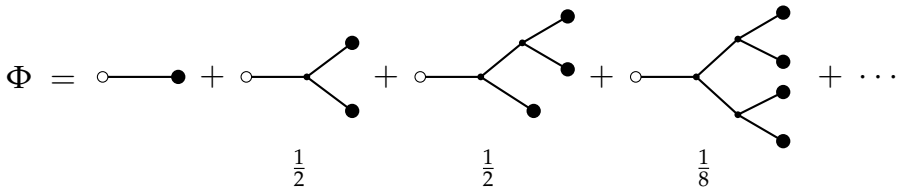


Figure 46: The classical field radiated by a source in the cubic theory, as the sum of tree diagrams with one external point (open dot). Large dots are the source and small dots the vertices; the numbers are the symmetry factors. Each iteration of (28.7) grafts new branches.

29 The effective action and the quantum vacuum★

The handwritten notes observe that the classical vacuum "receives quantum corrections" [hw. p. 7]. This chapter computes them. The tool is the effective action: an action whose classical theory reproduces the full quantum theory.

29.1 The Legendre transform

Let $\Phi(x) \equiv \frac{\delta}{i\,\delta J(x)} W[J] = \langle \phi(x) \rangle_J$ be the mean field in the presence of the source. Trading J for Φ defines the effective action:

The effective action

$$\frac{i}{\hbar}\,\Gamma[\Phi] = W[J] - i(J,\Phi), \qquad \frac{\delta\Gamma}{\delta\Phi(x)} = -\hbar J(x). \tag{29.1}$$

Without source the exact vacuum value $\Phi_0 = \langle \Omega | \hat{\phi} | \Omega \rangle$ is a stationary point of Γ, and at tree level $\Gamma = S$.

The second equation follows from the first: $\frac{i}{\hbar}\frac{\delta\Gamma}{\delta\Phi(x)} = \int \mathrm{d}^D y \left[\frac{\delta W}{\delta J(y)} - i\Phi(y)\right] \frac{\delta J(y)}{\delta\Phi(x)} - iJ(x) = -iJ(x)$, because $\delta W/\delta J = i\Phi$. At tree level $W = \frac{i}{\hbar}\big(S[\Phi_J] + \hbar(J,\Phi_J)\big)$ (§28.4), so $\Gamma = S$.

The definition has a direct reading. W is obtained from Γ exactly as W_{tree} is obtained from S: solve $\delta\Gamma/\delta\Phi = -\hbar J$ and evaluate $\frac{i}{\hbar}(\Gamma + \hbar(J,\Phi))$. So W is the sum of the tree diagrams of the "action" Γ, and the quantum theory is the classical theory of Γ. Differentiating $\delta\Gamma/\delta\Phi = -\hbar J$ once more, with $\delta\Phi(x)/\delta J(y) = i\,G_{\mathrm{c}}^{(2)}(x,y)$, gives

$$\frac{\delta^2\Gamma}{\delta\Phi(x)\,\delta\Phi(y)} = i\hbar\,\big(G_{\mathrm{c}}^{(2)}\big)^{-1}(x,y)\,: \tag{29.2}$$

the second derivative of Γ is the inverse of the full propagator. (At tree level $S^{(2)} = -(-\partial^2 + \mu^2)$ and $i\hbar\,G_{\mathrm{F}}^{-1} = -(-\partial^2+\mu^2)$, by (28.3).) Because every reducible structure is generated by the trees, the higher derivatives of Γ are the *one-particle-irreducible* vertices: the diagrams that stay connected when any single line is cut, with their external lines removed.

29.2 One loop from stationary phase

Expand the exponent of $\mathcal{Z}[J]$ about its stationary point ϕ_J, where $\delta S/\delta\phi = -\hbar J$. With $\phi = \phi_J + \zeta$,

$$\frac{i}{\hbar}\big(S[\phi] + \hbar(J,\phi)\big) = \frac{i}{\hbar}\big(S[\phi_J] + \hbar(J,\phi_J)\big) - \frac{i}{2\hbar}\int \mathrm{d}^D x\,\zeta\,A[\phi_J]\,\zeta + O(\zeta^3), \tag{29.3}$$

with the fluctuation operator $A[\phi] = -\partial^2 + U''(\phi) - i\eta$. The Gaussian integral over the fluctuations gives $\det^{-1/2} A[\phi_J]$, as for the oscillator of §9, where the propagator was a classical action times a fluctuation determinant. Hence $W = \frac{i}{\hbar}\big(S[\phi_J] + \hbar(J,\phi_J)\big) - \frac{1}{2}\operatorname{Tr}\ln A[\phi_J] + O(\hbar)$, up to a constant. In the Legendre transform $\Phi = \phi_J + O(\hbar)$, and the shift does not matter at this order because $S + \hbar(J,\cdot)$ is stationary at ϕ_J:

The one-loop effective action

$$\Gamma[\Phi] = S[\Phi] + \frac{i\hbar}{2}\operatorname{Tr}\ln\big(-\partial^2 + U''(\Phi) - i\eta\big) + O(\hbar^2), \tag{29.4}$$

$$\Gamma_{\rm E}[\Phi] = S_{\rm E}[\Phi] + \frac{\hbar}{2}\operatorname{Tr}\ln\big(-\bar\partial^2 + U''(\Phi)\big) + O(\hbar^2) \quad \text{(Euclidean)}. \tag{29.5}$$

29.3 The effective potential

For a constant Φ the Euclidean effective action is the volume of Euclidean spacetime times a function of Φ, the effective potential. With $S_{\rm E} = VT_{\rm E}\,U(\Phi)$ and $\operatorname{Tr}\ln(-\bar\partial^2 + M^2) = VT_{\rm E}\int\frac{{\rm d}^D\bar k}{(2\pi)^D}\ln(\bar k^2 + M^2)$:

The one-loop effective potential

$$V_{\rm eff}(\Phi) = U(\Phi) + \frac{\hbar}{2}\int\frac{{\rm d}^D\bar k}{(2\pi)^D}\,\ln\big(\bar k^2 + U''(\Phi)\big) + O(\hbar^2). \tag{29.6}$$

The one-loop term can be read in three ways.

1. **Zero-point energy in a background.** The $\bar k^0$-integral gives $\int\frac{{\rm d}\bar k^0}{2\pi}\ln(\bar k_0^2 + \omega^2) = \omega + \text{const}$ (Appendix E.5), so

$$V_{\rm eff}(\Phi) = U(\Phi) + \int\frac{{\rm d}^N k}{(2\pi)^N}\,\frac{1}{2}\hbar\,\omega_{\mathbf{k}}(\Phi), \qquad \omega_{\mathbf{k}}(\Phi)^2 = \mathbf{k}^2 + U''(\Phi). \tag{29.7}$$

   The fluctuations about a constant background are oscillators whose frequencies depend on the background, eq. (20.3). Their zero-point energy, which cancelled in §24.4, now depends on Φ, and that dependence is physical, as the dependence on a was in the Casimir effect.

2. **A closed worldline in a background.** With $\ln(\bar k^2 + M^2) = -\int_0^\infty\frac{{\rm d}\sigma}{\sigma}\,e^{-\sigma(\bar k^2 + M^2)} + \text{const}$,

$$V_{\rm eff}(\Phi) - U(\Phi) = -\frac{\hbar}{2}\int_0^\infty\frac{{\rm d}\sigma}{\sigma}\,(4\pi\sigma)^{-D/2}\,e^{-\sigma\,U''(\Phi)} + \text{const}. \tag{29.8}$$

   This is the closed worldline of eq. (17.10), with μ^2 replaced by $U''(\Phi)$: the worldline's mass is set by the field it moves through.

3. **Diagrams.** Expanding the logarithm in powers of $V''_{\text{int}}(\Phi)$ generates the one-loop one-particle-irreducible diagrams with any number of external legs at zero momentum: rings.

Check. In the quartic theory $U'' = \mu^2 + \frac{g_4}{2}\Phi^2$, and

$$\frac{\partial^2}{\partial\Phi^2}(V_{\text{eff}} - U)\Big|_{\Phi=0} = \frac{\hbar}{2}\int \frac{\mathrm{d}^D\bar{k}}{(2\pi)^D}\,\frac{g_4}{\bar{k}^2+\mu^2} = \frac{g_4}{2}\,G_{\text{E}}(0), \tag{29.9}$$

the mass shift (26.10) found from the tadpole diagram.

Curved spacetime. Replace $-\bar{\partial}^2$ by the Laplacian of a curved Euclidean metric. The same proper-time representation, now with the heat kernel of the curved-space operator, gives the one-loop effective action of a field in a gravitational background. Its expansion at small σ, the DeWitt–Schwinger expansion, produces a cosmological constant, the Einstein–Hilbert term and terms quadratic in the curvature, and it underlies the trace anomaly and semiclassical gravity. Part II's proper-time worldline is the tool that carries over.

29.4 The quantum vacuum and the tadpole

The true vacuum is a stationary point of Γ, so for a constant field it minimizes V_{eff}, not U [hw. p. 7]. In the cubic theory

$$\frac{\partial}{\partial\Phi}(V_{\text{eff}} - U)\Big|_{\Phi=0} = \frac{\hbar}{2}\int \frac{\mathrm{d}^D\bar{k}}{(2\pi)^D}\,\frac{g_3}{\bar{k}^2+\mu^2} = \frac{g_3}{2}\,G_{\text{E}}(0) \neq 0, \tag{29.10}$$

and the minimum moves to $\Phi_0 \approx -\frac{g_3}{2\mu^2}G_{\text{E}}(0)$, a shift of order $\hbar$. This is the tadpole (26.6) once more, found now from the potential. The identification of the vacuum with $\phi = 0$ [hw. p. 15] therefore fails at order $\hbar$. The remedy is to re-centre the field at the true minimum, or equivalently to add a linear term to the Lagrangian with a coefficient chosen to cancel the tadpoles order by order; then $\langle\Omega|\hat{\phi}|\Omega\rangle = 0$ and diagrams with tadpoles are dropped. A potential with only even powers has $\langle\phi\rangle = 0$ by the symmetry $\phi \to -\phi$, unless the symmetry is broken spontaneously (§20.6); then V_{eff} has two minima and one chooses between them.

Finally, the integral in (29.6) diverges at large $\bar{k}$ whenever $D \geq 2$. As written, the effective potential is infinite. Making sense of it is the subject of the next chapter.

30 Making sense of $G^{(N)}$: renormalization★

This chapter takes up Goal II [hw. p. 1]. It shows where the infinities come from, which theories can be rescued, how the rescue works in the simplest complete example, and what is left over: couplings that depend on the scale at which they are measured.

30.1 Where the infinities come from

A loop integral $\int \mathrm{d}^D k/(k^2)^n$ diverges at large k when $D \geq 2n$: it receives contributions from arbitrarily short wavelengths. The simplest example is the fluctuation at a point,

$$G_{\mathrm{E}}(0) = \hbar \int \frac{\mathrm{d}^D \bar{k}}{(2\pi)^D} \frac{1}{\bar{k}^2 + \mu^2} \overset{D=4}{=} \frac{\hbar}{16\pi^2}\Big[\Lambda^2 - \mu^2 \ln \frac{\Lambda^2}{\mu^2}\Big] + O(\Lambda^{-2}), \tag{30.1}$$

evaluated with a cutoff $|\bar{k}| < \Lambda$ (Appendix E.7). For $D \geq 2$ the field fluctuates infinitely at every point: it is too rough to have a value there, just as Brownian paths are too rough to have a velocity (eq. (4.14)). The physical reason is that the continuum has infinitely many degrees of freedom per unit volume, each with its own fluctuation. In quantum mechanics, $D = 1$, the same integrals converge, which is why the continuum limit of Part I caused no trouble.

To proceed one regulates: a cutoff Λ on Euclidean momenta, the lattice of §19.4 with $\Lambda \sim \pi/a$, or dimensional regularization, which continues the integrals in D. Results expressed in terms of measured quantities do not depend on the choice.

30.2 Power counting

At large loop momenta a diagram with L loops and I internal lines behaves like $\int \mathrm{d}^{DL} k/k^{2I}$, so its superficial degree of divergence is $\omega = DL - 2I$. In a theory with an n-leg vertex, $L = I - v + 1$ and $nv = 2I + E$, so

$$\omega = D - \frac{D-2}{2} E - \delta_n v, \qquad \delta_n = D - \frac{n(D-2)}{2}, \tag{30.2}$$

where δ_n is the mass dimension of g_n in units with $\hbar = c = 1$ (the effective coupling $\hat{g}_n$ of (20.10) has length dimension $-\delta_n$). Three cases follow.

- $\delta_n > 0$, *super-renormalizable*: ω decreases with v, so only finitely many diagrams diverge.
- $\delta_n = 0$, *renormalizable*: ω does not depend on v, so the same few Green's functions diverge at every order.
- $\delta_n < 0$, *non-renormalizable*: every Green's function diverges at high enough order.

theory	δ_n	ω	class
ϕ^3, $D=4$	1	$4-E-v$	super-renormalizable
ϕ^4, $D=4$	0	$4-E$	renormalizable
ϕ^3, $D=6$	0	$6-2E$	renormalizable
ϕ^6, $D=4$	-2	$4-E+2v$	non-renormalizable

For ϕ^4 in $D=4$ the divergent functions are $E=0$ (the vacuum energy, $\omega=4$), $E=2$ (the mass, $\omega=2$, together with a logarithmic divergence proportional to k^2 that renormalizes the normalization of the field from two loops on) and $E=4$ (the coupling, $\omega=0$). These are exactly the parameters already in the Lagrangian. Renormalizability means that these divergences can be absorbed into them.

30.3 The effective potential with a cutoff

Take the quartic theory in $D=4$ and write the potential with *bare* parameters, the ones that appear in the Lagrangian, including a constant vacuum energy:

$$U_{\rm B}(\phi)=\rho_{\rm B}+\frac{1}{2}\mu_{\rm B}^2\phi^2+\frac{g_{\rm B}}{4!}\phi^4,\qquad M^2(\phi)\equiv U_{\rm B}''(\phi)=\mu_{\rm B}^2+\frac{g_{\rm B}}{2}\phi^2. \tag{30.3}$$

With $|\bar{k}|<\Lambda$, the one-loop term of (29.6) is (Appendix E.7)

$$\frac{\hbar}{2}\int_{|\bar{k}|<\Lambda}\frac{\mathrm{d}^4\bar{k}}{(2\pi)^4}\,\ln\left(\bar{k}^2+M^2\right)=\frac{\hbar}{64\pi^2}\Big[2\Lambda^2M^2+M^4\Big(\ln\frac{M^2}{\Lambda^2}-\frac{1}{2}\Big)\Big]+\text{const}+O(\Lambda^{-2}). \tag{30.4}$$

The divergent pieces, $2\Lambda^2M^2$ and $-M^4\ln\Lambda^2$, are polynomials in ϕ of degree at most four, exactly the form of $U_{\rm B}$. Every infinity has the shape of a term already in the Lagrangian: this is renormalizability at its most concrete.

30.4 Renormalization: absorbing the cutoff

Choose a reference scale M_*, any mass, and define renormalized parameters by moving the divergent pieces into them. On the right, μ and g may be taken bare or renormalized, since the difference is of order $\hbar^2$:

$$\begin{aligned}\mu_{\rm R}^2&=\mu_{\rm B}^2+\frac{\hbar g}{32\pi^2}\Big[\Lambda^2-\mu^2\ln\frac{\Lambda^2}{M_*^2}\Big],\qquad g_{\rm R}=g_{\rm B}-\frac{3\hbar g^2}{32\pi^2}\ln\frac{\Lambda^2}{M_*^2},\\ \rho_{\rm R}&=\rho_{\rm B}+\frac{\hbar}{64\pi^2}\Big[2\Lambda^2\mu^2-\mu^4\ln\frac{\Lambda^2}{M_*^2}\Big].\end{aligned} \tag{30.5}$$

Expanding $M^4\ln(M^2/\Lambda^2)=M^4\ln(M^2/M_*^2)-M^4\ln(\Lambda^2/M_*^2)$ in (30.4) and collecting powers of ϕ:

The renormalized one-loop effective potential

$$V_{\text{eff}}(\phi) = \rho_{\text{R}} + \frac{1}{2}\mu_{\text{R}}^2\phi^2 + \frac{g_{\text{R}}}{4!}\phi^4 + \frac{\hbar}{64\pi^2}\,M^4(\phi)\Big[\ln\frac{M^2(\phi)}{M_*^2} - \frac{1}{2}\Big] + O(\hbar^2), \tag{30.6}$$

with $M^2(\phi) = \mu_{\text{R}}^2 + \frac{g_{\text{R}}}{2}\phi^2$. The cutoff has disappeared.

The bare parameters depend on Λ, and must, so that the predictions do not. The predictions, expressed through μ_{R}, g_{R} and ρ_{R}, which are fixed by measurements, are finite and independent of the cutoff. That is renormalization. It is not a device for hiding infinities. The parameters in the Lagrangian are not the quantities experiments measure, and the relation between the two depends on physics at short distances, which the cutoff stands in for.

- **Fine tuning.** By (30.5), $\mu_{\text{R}}^2 \approx \mu_{\text{B}}^2 + \hbar g\Lambda^2/32\pi^2$. To keep a scalar light, the bare mass must cancel a term of order Λ^2. This quadratic sensitivity is the hierarchy problem of the Higgs mass. The vacuum energy needs a cancellation of order Λ^4: the cosmological constant problem of §24.4 again.
- **Scheme dependence.** The constant $-\frac{1}{2}$ belongs to this choice of subtraction. Dimensional regularization with minimal subtraction gives $-\frac{3}{2}$ instead. The difference is a finite redefinition of the parameters and changes no prediction.

30.5 The running coupling

With g_{B} and Λ fixed, the renormalized coupling depends on the scale M_* at which it is defined. From (30.5),

$$\beta(g_{\text{R}}) \equiv \frac{\mathrm{d}g_{\text{R}}}{\mathrm{d}\ln M_*} = \frac{3\hbar\, g_{\text{R}}^2}{16\pi^2} + O(\hbar^2). \tag{30.7}$$

The same result comes from scattering. The amputated connected four-point function at one loop is the vertex plus a bubble with $S = 2$ in each of the three channels. With the momentum-space rules of §26.3 and a Wick rotation of the loop momentum,

$$\begin{gathered}\text{amputated } G_{\text{c}}^{(4)} = -\frac{i}{\hbar}\,g_{\text{eff}}, \qquad g_{\text{eff}} = g_{\text{B}} - \frac{\hbar g^2}{2}\sum_{c=s,t,u} I(p_c), \\ I(p) = \int \frac{\mathrm{d}^4\bar{\ell}}{(2\pi)^4}\,\frac{1}{(\bar{\ell}^2+\mu^2)\big((\bar{\ell}+p)^2+\mu^2\big)},\end{gathered} \tag{30.8}$$

where p_c is the Euclidean momentum flowing through channel c. With the cutoff, $I(p) = \frac{1}{16\pi^2}\int_0^1 \mathrm{d}x\,\big[\ln\frac{\Lambda^2}{\mu^2+x(1-x)p^2} - 1\big]$ (Appendix E.7), which becomes $\frac{1}{16\pi^2}\big[\ln\frac{\Lambda^2}{p^2} + 1\big]$ for $p^2 \gg \mu^2$. Define the coupling $g_{\text{R}}(M_*)$ as g_{eff} with all three channels at the scale

M_*. Then, with all three at a scale p,

$$g_{\text{eff}}(p) = g_{\text{R}}(M_*) + \frac{3\hbar g^2}{16\pi^2} \ln \frac{p}{M_*}. \tag{30.9}$$

The strength of the interaction measured in scattering at momentum p grows logarithmically with p, at the rate (30.7). Integrating the β-function,

$$\hbar g(p) = \frac{\hbar g(M)}{1 - \dfrac{3\hbar g(M)}{16\pi^2} \ln \dfrac{p}{M}}. \tag{30.10}$$

The coupling is weak at long distances, grows at short distances, and becomes singular at the *Landau pole* $p = M\, e^{16\pi^2/3\hbar g(M)}$ (Fig. 47), far beyond any experiment when the coupling is small. Physically, the vacuum fluctuations screen the interaction at long distances, as they screen electric charge in QED. If the cutoff is removed with the low-energy coupling held fixed, $g_{\text{R}} \to 0$: ϕ^4 theory in four dimensions is believed to be trivial, and it makes sense as an effective theory below a cutoff, which is how the Higgs sector of the Standard Model is understood. The opposite behaviour, a coupling that decreases at short distances, is asymptotic freedom: it occurs for ϕ^3 in $D = 6$ and for the strong interaction.

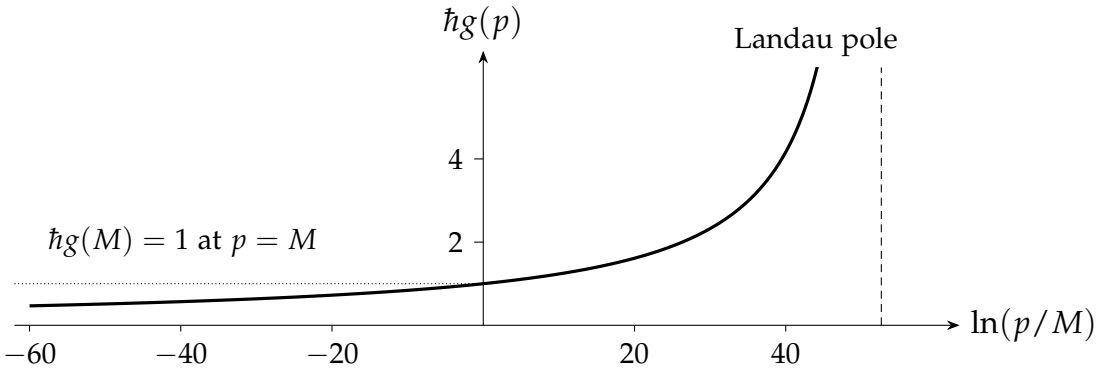


Figure 47: The running coupling of ϕ^4 theory, eq. (30.10), with $\hbar g = 1$ at $p = M$. It decreases logarithmically towards long distances and diverges at the Landau pole, $\ln(p/M) = 16\pi^2/3 \approx 52.6$.

30.6 Wilson's view

Take the cutoff seriously, as the lattice of §19.4. Integrate out the modes with $\Lambda/b < |k| < \Lambda$. The remaining modes see an effective action of the same form with changed couplings, plus new terms of higher dimension. Measured in units of the cutoff, a coupling of mass dimension δ is multiplied by b^δ, up to logarithms. Couplings with $\delta > 0$, such as μ^2, are *relevant*: they grow towards long distances. Those with $\delta = 0$,

such as g_4 in $D = 4$, are *marginal* and change logarithmically, as in (30.7). Those with $\delta < 0$, such as g_6, are *irrelevant* and die out like b^δ. At distances much longer than the cutoff only the relevant and marginal couplings survive. Any microscopic theory, seen from far away, therefore looks like a renormalizable one. This is the universality behind Part II's stone (§16.6), behind the critical exponents of statistical mechanics, and behind the polymer exponent of §25.4. Renormalizable theories are not singled out as fundamental; they are what any theory looks like at long distances.

Goal II in four sentences

Loop diagrams depend on the cutoff, because the continuum has infinitely many short-wavelength modes. In a renormalizable theory that dependence has the form of terms already in the Lagrangian. Expressed through measured parameters, the predictions are finite and independent of the cutoff. What remains is the dependence of the parameters on the scale at which they are measured: running couplings.

31 From Green's functions to experiment★

The handwritten notes treat the N-point functions as the definition of the theory [hw. p. 14]. They are not what an experiment measures. Experiments measure masses and scattering probabilities, and this chapter shows how both are read off from the $G^{(N)}$.

31.1 The spectral representation

Insert a complete set of states between the two fields of the Wightman function. Let λ label the states at rest, with invariant masses $\hbar\mu_\lambda$, and let $|\lambda_{\mathbf{k}}\rangle$ be their boosts, normalized as in Part II, $\langle\lambda_{\mathbf{k}}|\lambda'_{\mathbf{q}}\rangle = 2\omega_\lambda(\mathbf{k})\,\delta^N(\mathbf{k}-\mathbf{q})\,\delta_{\lambda\lambda'}$ with $\omega_\lambda(\mathbf{k}) = \sqrt{\mathbf{k}^2+\mu_\lambda^2}$. Translation invariance gives $\langle\Omega|\hat\phi(x)|\lambda_{\mathbf{k}}\rangle = \langle\Omega|\hat\phi(0)|\lambda_{\mathbf{k}}\rangle e^{ik\cdot x}$, and Lorentz invariance of the scalar $\hat\phi(0)$ makes the matrix element independent of $\mathbf{k}$. Comparing with the free Wightman function (15.4) of mass^2 s, D_s, gives (Appendix E.8)

The spectral (Källén–Lehmann) representation

$$\langle\Omega|\mathcal{T}\,\hat\phi(x)\hat\phi(y)|\Omega\rangle = \int_0^\infty \mathrm{d}s\;\rho(s)\;G_{\mathrm{F}}(x-y;s),$$
$$\tilde G^{(2)}(k) = \int_0^\infty \mathrm{d}s\;\rho(s)\;\frac{-i\hbar}{k^2+s-i\eta}, \tag{31.1}$$

$$\rho(s) = \frac{(2\pi)^N}{\hbar}\sum_\lambda \delta(s-\mu_\lambda^2)\,\left|\langle\Omega|\hat\phi(0)|\lambda_{\mathbf{0}}\rangle\right|^2 \;\geq 0, \tag{31.2}$$

where $G_{\mathrm{F}}(x;s)$ is the free Feynman function with μ^2 replaced by s (and $\langle\Omega|\hat\phi|\Omega\rangle = 0$).

For the free field only one-particle states contribute, with $\langle 0|\hat\phi(0)|\mathbf{k}\rangle = \sqrt{\hbar}/(2\pi)^{N/2}$ (eq. (15.11)), so $\rho(s) = \delta(s-\mu^2)$. In an interacting theory

$$\rho(s) = Z\,\delta(s-\mu_{\mathrm{p}}^2) + \rho_{\mathrm{cont}}(s), \tag{31.3}$$

an isolated one-particle pole of weight Z plus a continuum from the multiparticle states, which starts at $s = (2\mu_{\mathrm{p}})^2$ in the cubic theory and at $(3\mu_{\mathrm{p}})^2$ when the symmetry $\phi\to-\phi$ forbids the field to create two particles (Fig. 48). The canonical commutator gives the sum rule $\int \mathrm{d}s\,\rho(s) = 1$ (Appendix E.8), so $0 \leq Z \leq 1$: Z is the probability that the field creates a single particle.

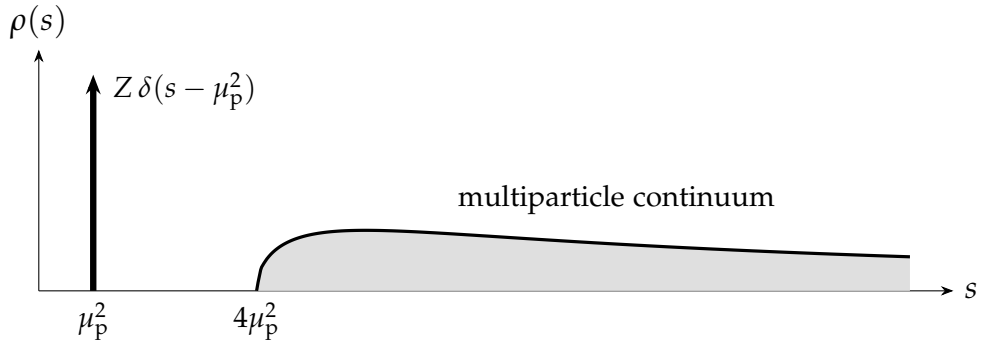


Figure 48: The spectral density of an interacting scalar (cubic theory): a one-particle pole of weight $Z \le 1$ at the physical mass, and a continuum of two-particle states from $s = 4\mu_{\mathrm{p}}^2$.

31.2 The physical mass is a pole

Near $k^2 = -\mu_{\mathrm{p}}^2$ the propagator is dominated by the one-particle term,

$$\tilde{G}^{(2)}(k) \approx \frac{-i\hbar\, Z}{k^2 + \mu_{\mathrm{p}}^2 - i\eta}. \tag{31.4}$$

The position of the pole is the mass of the particle, $m_{\mathrm{p}} = \hbar\mu_{\mathrm{p}}$, and its residue is Z. Beyond tree level the parameter μ in the Lagrangian is not the mass: in the quartic theory at one loop the self-energy is the tadpole, so $\mu_{\mathrm{p}}^2 = \mu^2 + \frac{g_4}{2} G_{\mathrm{E}}(0)$ by (26.10), a quadratically divergent shift. The natural renormalization condition is to set μ_{p} equal to the measured mass.

31.3 Scattering amplitudes: the LSZ formula

The creation operator of the free field can be extracted from $\hat{\phi}$ with the Klein–Gordon inner product (§15.6). In the interacting theory the same construction, divided by $\sqrt{Z}$, becomes the creation operator of an incoming particle as $t \to -\infty$ and of an outgoing one as $t \to +\infty$. Inserting it for every particle gives the formula of Lehmann, Symanzik and Zimmermann. With the normalization of these notes:

The LSZ reduction formula

$$\begin{aligned}\langle k_1' \cdots k_{n'}' | S | k_1 \cdots k_n \rangle = \prod_{\text{legs}} \frac{i}{(2\pi)^{N/2}\sqrt{\hbar Z}} \int \prod_j \mathrm{d}^D x_j\, e^{ik_j\cdot x_j} \big(-\partial_j^2 + \mu_{\mathrm{p}}^2 \big) \\ \times \prod_l \mathrm{d}^D x_l'\, e^{-ik_l'\cdot x_l'} \big(-\partial_l'^2 + \mu_{\mathrm{p}}^2 \big)\; G^{(n+n')}(x; x'),\end{aligned} \tag{31.5}$$

with all momenta on the mass shell.

In momentum space each operator $(-\partial^2 + \mu_{\rm p}^2)$ becomes $k^2 + \mu_{\rm p}^2$, which vanishes on shell and cancels the pole of the external propagator (31.4). Only terms with that pole on every leg survive: the *connected* diagrams, with their external propagators removed (amputated), and a factor $\sqrt{\hbar Z}$ for each leg in place of the propagator. Scattering amplitudes are the residues of the Green's functions at their one-particle poles. Removing the overall $(2\pi)^D \delta^D(\sum k)$ defines the invariant amplitude $\mathcal{M}$, in the standard normalization, by

$$i\mathcal{M} = \big(\sqrt{\hbar Z}\big)^{n+n'} \times \big(\text{sum of amputated connected diagrams}\big). \tag{31.6}$$

31.4 Two examples

Contact scattering in the quartic theory. At tree level the amputated diagram is the vertex, $-ig_4/\hbar$, so for $2 \to 2$ scattering

$$\mathcal{M} = -\hbar\, g_4, \tag{31.7}$$

a pure number in $D = 4$, as §20.5 said $\hbar g_4$ should be. With $\hbar = c = 1$ the differential cross-section in the centre-of-mass frame is $\mathrm{d}\sigma/\mathrm{d}\Omega = |\mathcal{M}|^2/64\pi^2 E_{\rm cm}^2$. At one loop g_4 is replaced by $g_{\rm eff}$ of (30.8): the running coupling of §30.5 is what scattering experiments at different energies measure.

Exchange in the cubic theory and the Yukawa potential. At tree level $2 \to 2$ scattering goes through one internal line in each of the three channels:

$$\mathcal{M} = \hbar\, g_3^2 \Big[\frac{1}{q_s^2 + \mu^2} + \frac{1}{q_t^2 + \mu^2} + \frac{1}{q_u^2 + \mu^2} \Big], \tag{31.8}$$

where q_c is the momentum carried by the exchanged particle. In the non-relativistic limit the t-channel term dominates and $q_t^2 \to |\mathbf{q}|^2$, the momentum transfer. The amplitude is then proportional to $1/(|\mathbf{q}|^2 + \mu^2)$, and in the Born approximation of non-relativistic quantum mechanics the potential is its Fourier transform, $V(r) \propto -e^{-\mu r}/r$: an attractive Yukawa potential, the one found from static sources in §24.3. The circle closes: from a field, to particles, to the force between them.

31.5 The volume in one page

Path integrals, Parts I–III

- **Part I.** $\langle x_f|e^{-i\hat{H}t/\hbar}|x_i\rangle = \int \mathcal{D}x\, e^{iS/\hbar}$: one particle, one time.
- **Part II.** $G_{\mathrm{F}}(x,y) = \int \frac{\mathcal{D}x\,\mathcal{D}\lambda}{V}\, e^{iS/\hbar} = \langle 0|\mathcal{T}\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle$: one worldline, summed over its proper length, equals the two-point function of a field.
- **Part III.** $G^{(N)} = \dfrac{\int \mathcal{D}\phi\, e^{iS/\hbar}\phi(x_1)\cdots\phi(x_N)}{\int \mathcal{D}\phi\, e^{iS/\hbar}} = \left(\dfrac{\delta}{i\delta J}\right)^N Z[J]\Big|_0$: a gas of worldlines that meet at vertices.

The same tools recur in all three parts.

- Stationary phase: the classical limit (§4.6), the on-shell modulus (§13.8), tree diagrams and the effective action (§28.4, §29.2).
- Gaussian integrals: the free particle and the oscillator (§9), the worldline (§13.3), the free field and Wick's theorem (§24.1).
- The $i\eta$ as a rotation of time: §8.9, §13.4, §23.4.
- Euclidean time as statistical mechanics: §8.7, §13.7, §23.5.
- Closed paths: the partition function (§8.11), the vacuum loop (§17.7), the zero-point energy and the effective potential (§24.4, §29.3).
- Contact terms: the Schwinger–Dyson equation (§7.7), the seagull (§17.5), the field hierarchy (§28.3).
- Gauge fixing: the worldline's Faddeev–Popov measure (§12.6). Gauge fields, where the same procedure needs ghost fields, lie beyond this volume.

E Solutions and supporting calculations

The handwritten notes leave several steps as exercises [hw. pp. 12, 15–17]. Their solutions come first, followed by the integrals used in the added chapters.

E.1 The functional Taylor expansion [hw. p. 12]

On a lattice with sites x_n and cell volume a^D the source is a list of numbers J_n, and Z is an ordinary function of them: $Z = \sum_N \frac{1}{N!} \sum_{n_1 \ldots n_N} \frac{\partial^N Z}{\partial J_{n_1} \cdots \partial J_{n_N}}\big|_0 J_{n_1} \cdots J_{n_N}$. The partial and functional derivatives are related by $\partial/\partial J_n = a^D \, \delta/\delta J(x_n)$, the lattice version of eq. (2.3), and $\sum_n a^D \to \int \mathrm{d}^D x$. Each of the N sums becomes an integral, and (23.11) follows. The coefficient $Z^{(N)}$ is symmetric in its arguments; the $1/N!$ compensates for the $N!$ orderings of the same set of points.

E.2 The free action in momentum space [hw. p. 15]

With $\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^{D/2}} e^{ik\cdot x} \tilde\phi(k)$ one has $\partial_\mu \phi = \int \frac{\mathrm{d}^D k}{(2\pi)^{D/2}} \, ik_\mu \, e^{ik\cdot x} \tilde\phi(k)$, and the x-integral gives $\int \mathrm{d}^D x \, e^{i(k+q)\cdot x} = (2\pi)^D \delta^D(k+q)$:

$$\int \mathrm{d}^D x \, \partial_\mu \phi \, \partial^\mu \phi = \int \mathrm{d}^D k \, \mathrm{d}^D q \, (-k\cdot q) \, \tilde\phi(k) \tilde\phi(q) \, \delta^D(k+q) = \int \mathrm{d}^D k \, k^2 \, \tilde\phi(k) \tilde\phi(-k). \quad \text{(E.1)}$$

In the same way $\int \mathrm{d}^D x \, \phi^2 = \int \mathrm{d}^D k \, \tilde\phi(k) \tilde\phi(-k)$ and $\int \mathrm{d}^D x \, J\phi = \int \mathrm{d}^D k \, \tilde J(k) \tilde\phi(-k)$, which gives (24.1).

E.3 Completing the square [hw. p. 16]

Write $A(k) = k^2 + \mu^2 - i\eta = A(-k)$ and substitute $\tilde\phi = \tilde\phi' + \hbar \tilde J / A$ into (24.1). The quadratic term gives $-\frac{1}{2} A \tilde\phi'(k) \tilde\phi'(-k) - \frac{\hbar}{2} \big[\tilde J(k) \tilde\phi'(-k) + \tilde\phi'(k) \tilde J(-k) \big] - \frac{\hbar^2}{2} \tilde J(k) \tilde J(-k)/A$, and the source term gives $\hbar \tilde J(k) \tilde\phi'(-k) + \hbar^2 \tilde J(k) \tilde J(-k)/A$. Under $\int \mathrm{d}^D k$ the two terms in the square bracket are equal (relabel $k \to -k$), so the terms linear in $\tilde\phi'$ cancel, and the terms in $\tilde J \tilde J$ add up to $+\frac{\hbar^2}{2} \tilde J(k) \tilde J(-k)/A$. This is (24.3).

E.4 Back to position space [hw. p. 17]

With $\tilde J(k) = \int \frac{\mathrm{d}^D x}{(2\pi)^{D/2}} e^{-ik\cdot x} J(x)$,

$$\frac{i\hbar}{2} \int \mathrm{d}^D k \, \frac{\tilde J(k) \tilde J(-k)}{A(k)} = \frac{i}{2} \int \mathrm{d}^D x \, \mathrm{d}^D y \, J(x) J(y) \, \hbar \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \frac{e^{-ik\cdot(x-y)}}{A(k)} = \frac{i}{2} (J, \Delta J), \quad \text{(E.2)}$$

using that Δ is even. This is (24.4).

E.5 The frequency integral of a logarithm

In real time the poles of $1/(-k_0^2+\omega^2-i\eta)$ lie at $k^0=\pm(\omega-i\eta')$. Closing the contour in the upper half-plane around $k^0=-\omega+i\eta'$, with residue $1/2\omega$, gives

$$\frac{\mathrm{d}}{\mathrm{d}\omega^2}\int\frac{\mathrm{d}k^0}{2\pi}\ln\big(-k_0^2+\omega^2-i\eta\big)=\int\frac{\mathrm{d}k^0}{2\pi}\,\frac{1}{-k_0^2+\omega^2-i\eta}=\frac{i}{2\omega}. \tag{E.3}$$

Integrating with $\mathrm{d}\omega^2=2\omega\,\mathrm{d}\omega$ gives $i\omega+\text{const}$. In Euclidean time $\int\frac{\mathrm{d}\bar{k}^0}{2\pi}\frac{1}{\bar{k}_0^2+\omega^2}=\frac{1}{2\omega}$, so $\int\frac{\mathrm{d}\bar{k}^0}{2\pi}\ln(\bar{k}_0^2+\omega^2)=\omega+\text{const}$. The constants are infinite but do not depend on ω.

E.6 The regulated Casimir sum

With $\omega_n=n\pi/a$ and $s=\pi/(a\Lambda)$,

$$\sum_{n\geq 1} n\,e^{-sn}=\frac{e^{-s}}{(1-e^{-s})^2}=\frac{1}{4\sinh^2(s/2)}=\frac{1}{s^2}-\frac{1}{12}+\frac{s^2}{240}-\cdots, \tag{E.4}$$

so $E(a)=\frac{\hbar\pi}{2a}\sum_n n\,e^{-sn}=\frac{\hbar\Lambda^2}{2\pi}a-\frac{\pi\hbar}{24a}+O\big(1/(a^3\Lambda^2)\big)$, which is (24.12).

E.7 One-loop integrals with a cutoff

In $D=4$, for a function of $\bar{k}^2$ only, $\int_{|\bar{k}|<\Lambda}\frac{\mathrm{d}^4\bar{k}}{(2\pi)^4}f(\bar{k}^2)=\frac{1}{16\pi^2}\int_0^{\Lambda^2}u\,f(u)\,\mathrm{d}u$, with $u=\bar{k}^2$; the angular volume is $2\pi^2$.

- **The fluctuation at a point.** $G_{\mathrm{E}}(0)=\frac{\hbar}{16\pi^2}\int_0^{\Lambda^2}\frac{u\,\mathrm{d}u}{u+\mu^2}=\frac{\hbar}{16\pi^2}\Big[\Lambda^2-\mu^2\ln\frac{\Lambda^2+\mu^2}{\mu^2}\Big]$, which gives (30.1).

- **The effective potential.** With $\int u\ln(u+M^2)\,\mathrm{d}u=\frac{u^2-M^4}{2}\ln(u+M^2)-\frac{u^2}{4}+\frac{M^2u}{2}$,

$$\begin{aligned}\frac{\hbar}{32\pi^2}\int_0^{\Lambda^2}u\ln(u+M^2)\,\mathrm{d}u=\frac{\hbar}{64\pi^2}\Big[2\Lambda^2M^2+M^4\Big(\ln\frac{M^2}{\Lambda^2}-\frac{1}{2}\Big)\Big]\\ +c(\Lambda)+O\Big(\frac{M^6}{\Lambda^2}\Big),\end{aligned} \tag{E.5}$$

  with $c(\Lambda)$ independent of M. This is (30.4).

- **The bubble.** Combine the denominators with a Feynman parameter, $\frac{1}{AB}=\int_0^1\frac{\mathrm{d}x}{[xA+(1-x)B]^2}$. For $A=(\bar{\ell}+p)^2+\mu^2$ and $B=\bar{\ell}^2+\mu^2$ the combination is $(\bar{\ell}+xp)^2+\Delta_x$ with $\Delta_x=\mu^2+x(1-x)p^2$. Shift $\bar{\ell}\to\bar{\ell}-xp$; with a cutoff this moves the boundary of the region by an amount that vanishes as $\Lambda\to\infty$. Then

$$I(p)=\frac{1}{16\pi^2}\int_0^1\mathrm{d}x\int_0^{\Lambda^2}\frac{u\,\mathrm{d}u}{(u+\Delta_x)^2}=\frac{1}{16\pi^2}\int_0^1\mathrm{d}x\,\Big[\ln\frac{\Lambda^2}{\Delta_x}-1\Big]+O(\Lambda^{-2}). \tag{E.6}$$

  For $p^2\gg\mu^2$, $\int_0^1\ln[x(1-x)]\,\mathrm{d}x=-2$ gives $I(p)=\frac{1}{16\pi^2}\big[\ln\frac{\Lambda^2}{p^2}+1\big]$.

E.8 The spectral representation and its sum rule

With the completeness relation $\mathbb{1} = |\Omega\rangle\langle\Omega| + \sum_\lambda \int \frac{\mathrm{d}^N k}{2\omega_\lambda(\mathbf{k})} |\lambda_\mathbf{k}\rangle\langle\lambda_\mathbf{k}|$ and $\langle\Omega|\hat\phi|\Omega\rangle = 0$,

$$\langle\Omega|\hat\phi(x)\hat\phi(y)|\Omega\rangle = \sum_\lambda \left|\langle\Omega|\hat\phi(0)|\lambda_\mathbf{0}\rangle\right|^2 \int \frac{\mathrm{d}^N k}{2\omega_\lambda(\mathbf{k})}\, e^{ik\cdot(x-y)}, \qquad k^0 = \omega_\lambda(\mathbf{k}). \tag{E.7}$$

The free Wightman function of mass^2 s is $D_s(x) = \int \frac{\mathrm{d}^N k}{(2\pi)^N} \frac{\hbar}{2\omega_\mathbf{k}(s)} e^{ik\cdot x}$ (eq. (15.4)), so the last integral is $\frac{(2\pi)^N}{\hbar} D_{\mu_\lambda^2}(x-y)$, and $\langle\Omega|\hat\phi(x)\hat\phi(y)|\Omega\rangle = \int \mathrm{d}s\, \rho(s) D_s(x-y)$ with ρ as in (31.2). Time ordering acts in the same way on both sides, which gives (31.1). For the sum rule, take the equal-time commutator $[\hat\phi(t,\mathbf{x}), \partial_t\hat\phi(t,\mathbf{y})] = i\hbar\,\delta^N(\mathbf{x}-\mathbf{y})$, valid when the interactions contain no derivatives, so that $\hat\Pi = \partial_t\hat\phi$. Each free D_s reproduces $i\hbar\,\delta^N$, so the representation gives $i\hbar\,\delta^N \int \mathrm{d}s\,\rho(s)$, and therefore $\int \mathrm{d}s\,\rho(s) = 1$.

F Map of the handwritten notes and correction log

F.1 Where each handwritten page went

pages	content	here
1	the problem; goals I and II; Part I's exact statement	§19.1
2	$x \leftrightarrow \tilde{\phi}(\mathbf{k})$; field eigenstates; Fourier transform	§19.2
3	the field path integral; the lattice	§19.2–19.4
4	comments: $i\epsilon$, classical solution, Lorentz invariance, vacuum	§20.1–20.2
5	choosing a vacuum; shifted field and potential	§20.3
6	boundary conditions; Taylor expansion of V; two challenges	§20.3–20.4
7	expansion in couplings and $\hbar$; quantum corrections to the vacuum	§20.4–20.6
8	$G_{if}^{(N)}$; the guess $G^{(N)} = \langle\Omega\|\mathcal{T}\cdots\|\Omega\rangle$	§23.1
9–10	spectral decomposition; Wick rotation; the vacuum dominates	§23.2
11	the ratio formula	§23.3
12	independence of the boundary conditions; $W[J]$; Taylor expansion	§23.3, §23.6
13–14	derivatives of the generating functional; recap and claim	§23.6
15	validity; the Lagrangian; free theory in momentum space	§19.3, §20.4, §24.1
16–17	completing the square; $W_0[J]$ and Δ; dumbbells	§24.1–24.2
18–19	rules for W_0; symmetry factors; $G^{(2)}$, $G^{(4)}$	§24.2
20–21	the interacting theory; $\tilde{W}[J]$; $F[\delta/i\delta J]$	§26.1
22	the double expansion; vacuum diagrams	§26.2, §27.2
23	examples of vacuum diagrams; bubbles factor out	§26.4, §27.2
24–25	connected and disconnected pieces; exponentiation	§27.1

F.2 Corrections

p. 2 $\mathrm{d}p$ is $\mathrm{d}^N k$. The operators $\hat{\tilde{\phi}}(\mathbf{k})$ are not Hermitian; their eigenvalues obey $\tilde{\phi}(-\mathbf{k}) = \tilde{\phi}(\mathbf{k})^*$ (§19.2).

p. 4 "Lorentz invariant only if constant" is false: $f(x^2)$ is Lorentz invariant. Translation invariance, or minimal energy, forces a constant (§20.2). A local minimum that is not global is a false vacuum; at a maximum the long-wavelength modes are tachyonic (§20.2, §20.6).

p. 5 The shifted Lagrangian has no $\frac{1}{2}$ in front of $V(\chi)$ or $U(\phi_0)$ (§20.3).

p. 6 m, λ_3, λ_4 become $\mu = m/\hbar$, g_3, g_4; μ^2 is the curvature of U at the vacuum (§20.4).

p. 7 The "power series in both the λ's and $\hbar$" is a single expansion, in $\hat{g}_n = \hbar^{n/2-1} g_n$ (§20.5).

pp. 9–10 $E_\Omega = 0$ and "a bounded system" are not needed. What is needed is a spectrum bounded below with a gap; the tilt $t \to \pm\infty(1 - i\eta)$ is essential; the overlaps with the vacuum never vanish (§23.2).

p. 11 $\Psi_\Omega[\phi_i]$ needs its complex conjugate. Both numerator and denominator carry $e^{-iE_\Omega(t_f - t_i)/\hbar}$, which cancels in the ratio (§23.3).

p. 13 The boxed $\delta W/\delta J$ omits the normalizing denominator (§23.6).

p. 15 The interaction terms of $\mathcal{L}$ have the wrong sign (§20.4). $|\Omega\rangle \leftrightarrow \phi = 0$ holds at tree level only (§29.4). $\tilde{J}(p)\phi(-p)$ should read $\tilde{J}(k)\tilde{\phi}(-k)$ (§24.1).

p. 17 In $T_0^{(4)}$ the first factor has iJ_x, not $-iJ_x$ (§24.2). The $i\epsilon$ is required, not optional (§24.1).

pp. 20, 22 V denotes the interaction and the number of vertices; here V_{int} and v.

p. 22 "Expanding around $\phi = 0$" means expanding in the couplings and the sources (§26.2).

p. 23 The cancellation of vacuum bubbles is a consequence of the exponentiation of pp. 24–25 and is proved after it (§27.2).

p. 24 The set of connected diagrams with a given number of sources is finite at each order in the couplings, not in total (§27.1).

notation $W \leftrightarrow Z$ swapped (generating functionals of all and of connected diagrams); $\lambda_n \to g_n$; $p \to k$; $\langle f \rangle \to \int \mathrm{d}^D x\, f$ and $\langle J\phi \rangle \to (J, \phi)$; Φ (p. 4) $\to \phi_{\mathrm{c}}$; $i\epsilon \to i\eta$.

In Part II, eq. (17.12) gives the connected part of $G^{(3)}$. At the same order in g_3 the full $G^{(3)}$ also contains the products of a line and a tadpole (§26.4); Part II has been amended accordingly.